How Institutional Structure Shapes Groundwater Market Implementation: Evidence from the Oxnard Basin

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Abstract

Groundwater markets are usually introduced on top of existing institutions, so what a market adds depends on what those institutions already do. This paper studies the Oxnard Basin in Ventura County, California, where the first groundwater market under California's Sustainable Groundwater Management Act was implemented. In the Oxnard Basin, operators may pool the allocations of multiple wells under a single reporting coalition and reallocate water internally with the absence of the formal market. Using a field-year panel of well-level extraction records linked to crop and land-use data, I show that these coalitions have become more concentrated, sort on crop perishability, and actively reallocate water among their members. Marginal values of groundwater recovered from a structural model, however, show no evidence that this reallocation moves water toward higher-value uses. Counterfactual regimes show that even coalitions that equalized marginal values perfectly would capture only 26 percent of the gains from a frictionless well-level market. Opening a market to wells outside coalitions raises this share to 63 percent, and the remaining 37 percent requires wells within coalitions to join the trade as individual agents. Because coalitions may pursue objectives other than equalizing marginal values of groundwater, the coalition boundary can act as a barrier to the formal market and not as a substitute for it. A rule prohibiting net purchases in environmentally vulnerable areas, which a formal market can enforce but internal reallocation cannot, retains 92 percent of the market's gain. The results imply that the definition of the trading entity is a first-order choice in groundwater market design, and a market built on an institution as complex as coalition-based operation needs careful implementation, because its design determines which gains are reachable and who participates.

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1 Introduction

Environmental markets have emerged as a popular market-based policy instrument for managing natural resources and pollution. By allowing permits to be traded among users with heterogeneous valuations, a cap-and-trade system can achieve aggregate resource-reduction targets and allocate resources to the highest-value users. The logic has motivated the adoption of tradable permit systems across a wide range of settings, from greenhouse gas emissions (Newell and Stavins, 2003; Fowlie, S. P. Holland, and Mansur, 2012) to fisheries (Newell, Sanchirico, and Kerr, 2005; C. Costello et al., 2016) to water (Rafey, 2023; Bruno and Jessoe, 2024). In the context of groundwater, where chronic overdraft threatens agricultural economies and ecosystems across the western United States, environmental markets have attracted growing interest as a tool for achieving sustainability targets while minimizing social costs (Ayres, Meng, and Plantinga, 2021; Bruno and Sexton, 2020). California's Sustainable Groundwater Management Act (SGMA), signed into law in 2014, accelerated this interest by requiring local Groundwater Sustainability Agencies (GSAs) to bring overdrafted basins into balance, thereby creating a binding regulatory framework for the implementation of groundwater markets.

Markets, however, are not free to build. The design of an effective groundwater market requires resolving a range of institutional questions that go beyond simply setting a cap and allowing trade: who is the unit of trade, how are initial allocations defined, what monitoring and enforcement infrastructure is required, and how do existing organizational relationships among users shape their incentives to participate (Bruno and Jessoe, 2024; Bruno, Hagerty, and Wardle, 2022)? Where groundwater markets have been implemented, participation and trading volumes have often fallen short of expectations (Hagerty, 2019). Deciding whether a market is worth its costs therefore requires evaluating what it adds, and that evaluation depends on the counterfactual. Existing studies typically compare a market with command-and-control regulation (Fowlie, S. P. Holland, and Mansur, 2012; Bruno and Sexton, 2020) or with open access (Ayres, Meng, and Plantinga, 2021). In practice, however, markets are rarely introduced into an institutional vacuum. They are layered on top of existing allocation systems, and users have often already found ways to move the resource among themselves under those systems. In fisheries, for example, collective arrangements such as sectors and the consolidation of quota among firms have developed alongside catch share programs (Huang et al., 2018; Yandle and Dewees, 2008). The relevant counterfactual for a new market is therefore the institution it builds on. If that institution already reallocates the resource efficiently, thin trading on the new platform is benign. If it does not, how the market is designed relative to the existing institution determines how much of the potential gain is realized.

This paper studies this question in the first groundwater market implemented under SGMA, established by the Fox Canyon Groundwater Management Agency (FCGMA) in the Oxnard Basin

of Ventura County, California, with trading beginning in 2020 (Heard, Fienup, and Remson, 2021). The Oxnard Basin is one of the most productive agricultural regions in Southern California, producing a dense mix of high-value perishable crops such as strawberries, raspberries, avocados, and specialty vegetables. The basin was designated critically overdrafted under SGMA, requiring a reduction in groundwater extraction to the sustainable level. The FCGMA, which had operated an allocation-based regulatory system since the 1990s, implemented the formal market to allow agricultural users to trade their groundwater allocations, improving allocative efficiency while achieving the basin-wide extraction target. Users face binding extraction constraints, crops differ in their water productivity, and the infrastructure for monitoring and reporting was already in place. Yet trading volumes have been strikingly low. Standard explanations for thin markets, such as transaction costs (Crase, Dollery, and Lockwood, 2002), market power (Bruno and Sexton, 2020), and lack of trust (Breetz et al., 2005), offer an incomplete account of this case, because the FCGMA operates a centralized, double-blinded trading platform, maintains detailed records of all allocations, and actively encourages participation.

The pre-market institution offers a clue. In a 2023 FCGMA Board Meeting, a GSA Board Member observed that large operators had effectively created "private markets" outside the formal trading platform, consolidating wells under shared operator codes and transferring water internally without any formal transaction.[1] The channel is the FCGMA's combined-code (CombCode, I will call this as "coalition" in the later paper) reporting system. Although allocations are assigned to individual wells, the FCGMA allows operators controlling multiple wells to pool their allocations under a single reporting unit and comply at the aggregate level. An operator can therefore extract more than the allocation assigned to one well, as long as extraction at another well in the same coalition is correspondingly reduced, effectively transferring allocation between wells without engaging in any formal market transaction. This structure creates a system of partial "unitization" (Kaffine and C. J. Costello, 2010), in which groups of wells are managed as a single unit for regulatory purposes. This observation motivates the central questions of this paper. Which wells form these coalitions? Do coalitions reallocate water the way a market would, by moving it toward higher-value uses? And what do the answers imply for the design of a formal market?

There are reasons to expect that coalitions may not behave like markets for water. As emphasized by Libecap (1990), differences in the distribution of costs and benefits among group members can raise the transaction costs of reaching an agreement, and members who expect unequal benefits may resist coordination even when it generates aggregate gains (Ayres, E. C. Edwards, and Libecap, 2018). Coordination is therefore easier among homogeneous users. In the Oxnard Basin, such groups exist within agricultural supply chains. The basin's dominant crops, such as berries

[1] https://water.ca.gov/-/media/CWC-Website/Files/Documents/2023/06_June/June2023

and specialty vegetables, are usually produced under long-term contractual relationships between growers and downstream packer-shippers, who coordinate planting schedules, delivery timing, and quality standards across multiple grower operations (Mighell and Jones, 1963; Martinez, 2002). These vertically coordinated relationships lower the cost of forming a groundwater coalition. But coalitions formed along supply chains are organized around the output market rather than around differences in the value of water, and their members tend to grow the same crops. Both features may limit how much efficiency gain a coalition can deliver, and whether it pursues that gain at all.

This paper provides four sets of findings on these questions. First, using well-level extraction records from the FCGMA spanning 2014-2023, linked to farm boundaries and crop characteristics through geospatial matching, I document the pattern of coalition formation. Coalitions have become increasingly concentrated over time, with the top five coalitions accounting for 61 percent of total agricultural groundwater extraction by 2023, up from 36 percent in 2014. Fields growing more perishable crops are more likely to belong to multi-well coalitions, and coalitions are significantly more crop-homogeneous than random groupings of fields. An event study exploiting the timing of wells transitioning from single well to a member of a coalition shows a sharp and persistent increase of about 1.06 log points in extraction at the joining well, indicating that coalitions actively reallocate water among their members.

Second, I study the reallocation mechanism using a structural model that recovers the marginal value of groundwater at each well from crop revenue, irrigation efficiency, and pumping cost. If coalitions reallocate water the way a market would, the wells that extract beyond their own allocation should be those with the highest marginal value at the allocation. I find no evidence of this. Within the same coalition-year, these "buyer" wells do not have higher marginal values than their partners, and a decomposition shows that, if anything, buyers have lower irrigation efficiency. Marginal values are more similar within coalitions than across the basin, but this similarity is largely explained by members growing the same crops. Evaluated in the model, the observed pattern of extraction within coalitions lowers profit relative to simply following well-level allocations, and spreads marginal values further apart rather than bringing them together. Coalitions reallocate water, but their reallocation appears to be driven by considerations other than the marginal value of groundwater. One explanation consistent with this pattern is that coalitions organized around supply contracts allocate water to secure committed output from each field rather than to maximize the value of groundwater.

Third, I use the recovered marginal values to solve for counterfactual market regimes under alternative definitions of who may trade with whom. Even if coalitions equalized marginal values perfectly among their members, they would capture only 26 percent of the gain from a frictionless, basin-wide well-level market. The limit is structural: because members grow similar crops, most of the dispersion in marginal values lies between coalitions rather than within them. Giving wells

outside coalitions access to a common market raises the captured share to 63 percent, and the remaining 37 percent can be realized only if coalition wells themselves trade in the formal market. A formal market that lets coalitions keep their allocations internally would therefore leave a large share of the potential gain unrealized, even under the most favorable assumption about the coalition behavior.

Fourth, I show that a formal market can address an external cost that internal reallocation cannot. Because the coalition system evaluates compliance only at the operator level, it cannot prevent an operator from shifting extraction toward wells in areas already vulnerable to seawater intrusion. A formal market can impose a directional rule at the point of trade. Prohibiting net purchases in the basin's two Special Protection Areas would foreclose about one-third of the reallocation that coalitions could carry out internally, but would cost a basin-wide market only 7.6 percent of its efficiency gain.

For policymakers designing groundwater markets under SGMA, the findings of this paper have a concrete implication. The definition of the transaction entity, i.e., the level at which allocations are assigned and at which compliance is evaluated, determines what the pre-market institution can do and how much is left for the market. Pooled compliance at the operator level does not substitute for a formal market: it reallocates water without targeting its value, it is limited by the crop homogeneity of the groups that form, and it cannot enforce directional restrictions. The design principle underlying the coalition (CombCode) system is not unique to Oxnard: analogous structures appear in the Mid-Kaweah and Madera subbasins, suggesting that the issues identified here may arise broadly across SGMA implementation.

This paper connects to three strands of literature. The first is the literature on collective action and the governance of common-pool resources (CPRs). After studying many cases of CPR management, Ostrom (2015) suggests that the resource users could themselves form a self-organized authority and manage the resource more effectively in the absence of a centralized government if certain institutional and social conditions are met. Libecap (1990) examines CPR governance through the lens of property rights negotiations, and he frames this as also a collective action problem that can be addressed through formal contractual agreements. While in Ostrom's theory, collective actions are rooted in social norms and cohesion, Libecap states that informal institutions rely on clearly designed property rights and negotiations. However, both highlight that groups with larger size and more heterogeneous attributes tend to face more challenges in collectively acting, and informational barriers can limit their scope (E. Edwards and Leonard, 2025). The paper also connects the unitization literature from oil and gas economics to groundwater governance (Kaffine and C. J. Costello, 2010), showing that partial unitization through pooled compliance can enable reallocation among users without directing the resource toward its highest-value uses.

Despite the richness of work grounded in empirical cases in regions with relatively weak central

governance (Libecap, 1989; Agrawal and Goyal, 2001; Adhikari and Lovett, 2006; Ruttan, 2008), studies of collective action in CPRs remain relatively rare in the settings of developed countries where formal regulatory institutions and property rights are better established. In such contexts, governance tends to rely heavily on top-down mechanisms, with less attention paid to the informal arrangements and coordination that may occur among users within those systems. There are, however, a few exceptions. Lopez-Gunn (2003) study the example of the Groundwater User Associations in La Mancha aquifers, Spain, to explore the characteristics that guarantee the success of self-regulation and the complementary relationship between self-regulating water management and higher-level authorities. Shalsi et al. (2022) use the Angas Bremer irrigation district located in South Australia to show that the local collective action can lead to successful groundwater management even under the formal institutions.

Related insights on how collective action works under the management of a higher-level, more centralized institution can also be found in other fields of natural resources. In the case of fisheries, researchers have studied community-based fishery management (Copes and Charles, 2004; Wiber et al., 2004), and the development of collective right-based management - "sectors"[2] - in New England. Sector management shows how collective action among fishermen under a catch share system can change their effort in fishing and shift the production frontiers (Huang et al., 2018). Similarly, under the Quota Management System (QMS) in New Zealand, transferable catch allowances are allocated to New Zealand-owned fishing companies. The voluntary consolidation of quotas among firms resembles forms of collective action and bargaining (Yandle and Dewees, 2008). In the field of pollution management, Yoder (2019) also found that a shared pollution cap incentivized farmers in the Florida Everglades to cooperate and improve the water quality. A common premise in this literature is that collective action, where it emerges, improves the management of the resource. This paper shows that collective action organized around another objective, here the coordination of perishable crop supply chains, can move a resource among group members without improving its allocation.

The second strand is the literature on resource management and environmental market design. Nowadays, some form of environmental market has been successfully established in practice. For example, individual transferable quotas (ITQs) are used in fisheries (Newell, Sanchirico, and Kerr, 2005; C. Costello et al., 2016; D. Holland, 2016), and cap-and-trade programs are employed as an instrument to regulate greenhouse gas emissions (Newell and Stavins, 2003; Fowlie, S. P. Holland, and Mansur, 2012).

While groundwater shares some institutional and economic parallels with other CPRs, such as fisheries and air pollution control, it also presents several unique challenges based on its context.

[2] https://www.fisheries.noaa.gov/new-england-mid-atlantic/commercial-fishing/sector-management-northeast-multispecies-fishery

As Bruno and Jessoe (2024) discusses, lack of monitoring and enforcement for non-compliance, non-uniform marginal damages from groundwater extraction, disconnected hydrology that limits coordinated management across regions, high costs of transactions and obtaining information, and temporal dynamics of aquifer conditions all create challenges for policy design.

A large body of groundwater-relevant literature simulates the substantial gains from the introduction of market-based approaches to manage groundwater. For example, Kuwayama and Brozović (2013) uses data on irrigation wells in Nebraska to find that adopting spatially differentiated groundwater extraction rules allows sizable savings in total social costs. Palazzo and Brozović (2014), also studying Nebraska, find that groundwater trading reduces the costs of agricultural water use restrictions, highlighting the importance of institutional contexts and initial permit design, and show that initially unused permits create incentives for trade. Guilfoos, Khanna, and Peterson (2016) test a multi-cell hydrological model on one aquifer located in northwest Kansas to find that simple, uniform pricing, quota, or environmental market policies on groundwater management perform poorly unless improved by more localized policies. Bruno and Sexton (2020) apply an agricultural groundwater trade model to southern California and estimate that the benefits of allowing trade are up to 36% greater than the command-and-control regime, even with potential market power existing.

Despite the rich studies on modeling and simulating the performance of policies, market-based approaches regulating groundwater extraction are rarely seen in practice. There are a few exceptions. For example, Fishman et al. (2016) shows field evidence from a designed program implemented in Gujarat, India, to suggest that shifting to meter-based billing did not substantially change groundwater use. Smith et al. (2017) utilize a difference-in-difference framework to estimate that the self-imposed groundwater extraction fee in San Luis Valley, Colorado, had reduced the groundwater use by 33%. Ayres, Meng, and Plantinga (2021) looked at the groundwater market in Mojave, California, and used a spatial regression discontinuity model to estimate that the environmental market increased land value by half compared to the open access regime.

The third strand is the literature on vertical coordination in agricultural production. Following Mighell and Jones (1963), economists have documented the prevalence of vertical coordination in agricultural supply chains and examined the efficiency and distributional consequences of vertical integration and contracting. Several works studied the crop attributes, such as high value and perishability, in determining the vertically coordinated structure (Birthal, Joshi, and Gulati, 2005; Lo, 2010). The literature has largely focused on output markets and production decisions. The implications of vertical coordination for the management of production inputs, particularly the natural resources, have received less attention. By documenting how vertically coordinated supply chains shape which growers pool their groundwater, and showing that the resulting reallocation does not follow the value of water, this paper highlights a channel through which organization in

the output market shapes the allocation of a natural resource input.

This paper makes three contributions. First, it provides the first empirical study of a groundwater market implemented under SGMA that takes consideration of the pre-market institution, documenting which users pool their allocations and testing whether their reallocation targets the value of water. Second, it develops a structural framework that recovers well-level marginal values of groundwater and uses them both to test the reallocation mechanism within existing groups and to decompose the gains from a formal market across alternative definitions of the trading entity. The framework applies to other cap-and-trade settings where compliance is evaluated for groups rather than individual units. Third, it shows that the definition of the regulated entity determines how much efficiency gain a market can deliver, and which external costs the regulator is able to control. Taken together, these contributions suggest that effective environmental market design requires attention not only to the cap level and the trading rules, but to the organizational boundaries within which compliance is evaluated.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, including the FCGMA regulatory framework, the Oxnard Basin's agricultural production structure, and the coalition system. Section 3 describes the data sources and the construction of the field-level panel. Section 4 presents the reduced-form evidence on coalition formation and its effect on extraction. Section 5 develops the conceptual model, and Section 6 estimates the marginal value of groundwater and tests whether reallocation within coalitions follows it. Section 7 solves for counterfactual market equilibria under alternative trading-group definitions, compares them with observed coalition behavior, and evaluates a directional trading restriction. Section 8 concludes with policy implications for groundwater market design under SGMA.

2 Institutional Background & Setting

2.1 Agricultural Production in Oxnard Basin

The Oxnard Basin is one of the most productive agricultural regions in Ventura County, with irrigation accounting for approximately 60% of total groundwater use. The region's top cash crops include strawberries, raspberries, nursery stock, and some more water-intensive crops, such as avocado and leafy vegetables[3].

The crop mix reflects significant implications for the structure of agricultural production in the basin. Strawberries alone typically account for the largest share of harvested acreage and farm revenue in Ventura County, which ranks consistently among the top ten California counties by

[3]Groundwater Sustainability Plan for the Oxnard Subbasin. https://cawaterlibrary.net/document/groundwater-sustainability-plan-for-the-oxnard-subbasin/

total agricultural production value. Most leading commodities are highly perishable, with short post-harvest shelf lives that create strong incentives for close coordination between growers and downstream handlers, processors, and marketers. The economics of perishable crop production, for example, high revenue per acre, tight timing requirements, and significant exposure to weather and market risk, have incentivized the vertically integrated and contractually organized supply chains.

The scale and structure of farming operations differ from the family-farm model often associated with California agriculture. A significant share of production is concentrated among commercial growers managing multiple fields across the basin under contractual relationships with large packer-shippers. This concentration, combined with the organizational demands of perishable crop supply chains, means that growers in he Oxnard Basin are connected through dense contractual and operational relationships, which is a feature with important consequences for how groundwater is managed in practice.

2.2 The FCGMA Regulatory Framework

The Oxnard Basin spans approximately 58,000 acres in southern Ventura County, with an onshore storage capacity of about 7,140,000 acre-feet and average extraction of roughly 69,000 acre-feet per year between 2015 and 2017[4][5].Groundwater supplies about 67% of total water demand and about 85% of total irrigation demand, with the remainder met through surface water and recycled sources.

Groundwater management in the Oxnard Basin falls under the Fox Canyon Groundwater Management Agency (FCGMA), established in 1983 under the California Assembly Bill 2995. The FCGMA is one of the earliest binding groundwater allocation agencies in California. It has implemented an individual extraction allocation system assigning annual pumping caps to each well. Ordinance No. 5 (1990) established this structure: wells with extraction records from 1985 to 1989 received allocations starting at 95% of their baseline average, declining incrementally to 75%, while post-1989 wells were allocated based on crop type, irrigated acreage, and irrigation efficiency estimates. Excess extraction triggers a tiered surcharge. The ordinance targeted a basin-wide reduction to a safe yield of 120,000 acre-feet per year by 2010.

Regulatory pressure intensified significantly with the passage of California's Sustainable Groundwater Management Act (SGMA) in 2014, which established a statewide framework requiring local groundwater sustainability agencies to bring critically overdrafted basins into balance. The Oxnard

[4]The offshore storage capacity is approximately 3,360,000 acre-feet for all aquifers in the Oxnard Basin, but intrusion of saline water has been observed in some areas, such as the Port Hueneme and Point Mugu.

[5]https://water.ca.gov/-/media/DWR-Website/Web-Pages/Programs/Groundwater-Management/Bulletin-118/Files/2003-Basin-Descriptions/4_004_02_OxnardSubbasin.pdf

Basin was among the basins designated critically overdrafted under SGMA, a designation reflecting its history of chronic overdraft and vulnerability to seawater intrusion.

In response, the FCGMA designed a groundwater market in 2019 and implemented it in 2020. This market was the first formal groundwater market established under SGMA in California (Heard, Fienup, and Remson, 2021). The market was intended to allow agricultural users to trade groundwater allocation credits, reducing extraction costs and improving allocative efficiency among users with heterogeneous water needs. Despite this regulatory setup, observed trading volumes remained low. At the same time, evidence of informal reallocation emerged through official channels. At a 2023 FCGMA Board Meeting, a GSA Board Member observed that large operators had effectively created "private markets" outside the formal trading platform, consolidating wells under shared operator codes and transferring water internally.[6] This observation motivates the focus of this paper on the pre-market institution: what these internal "markets" look like, how they reallocate water, and what they imply for the design of the formal market.

2.3 Well Consolidation and CombCode Structure

A key institutional feature of the FCGMA's regulatory structure is the combined-code (CombCode, I refer them as "coalition" througout the paper) reporting system. While allocations are assigned at the individual well level, compliance is evaluated at the operator level. The FCGMA defines an operator as a person or entity who owns or operates a groundwater extraction facility with the written approval of the owner (FCGMA Ordinance No. 5). An operator may oversee multiple wells, each with its own allocation, but the FCGMA evaluates compliance by comparing aggregate extraction across all wells sharing the same CombCode. Ordinance No. 5 explicitly states that "where an operator operates more than one extraction facility, the extraction allocations for the individual facilities may be combined."

This provision has an important practical consequence: operators controlling multiple wells need only comply with the sum of allocations across their portfolio, not the allocation assigned to any individual well. An operator can therefore extract more than a single well's allocation as long as extraction elsewhere in the portfolio is correspondingly reduced — effectively transferring allocation between wells without any formal market transaction. This internal flexibility closely mirrors the logic of a cap-and-trade system but operates entirely within the operator's organizational boundary.

Any change in operator status, including consolidation of wells under the same coalition, must be formally approved by the FCGMA prior to the reporting period in which it takes effect. This

[6] https://water.ca.gov/-/media/CWC-Website/Files/Documents/2023/06_June/June2023

limits strategic manipulation of operator assignments and ensures that observed coalition reflect organizational relationships.

This institutional arrangement generates the core empirical variation exploited in this paper. Coalition assignments define the boundaries within which internal reallocation can occur. Because those assignments are observed directly in FCGMA extraction records, the data allow me to identify coordination groups, track their evolution, and compare extraction behavior within and across groups.

The supply chain relationships described above map naturally onto this institutional structure. Operators who already coordinate production and logistics across multiple fields face low additional costs of consolidating wells under the same coalition. Whether the resulting internal flexibility is used to move water toward its most valuable uses, however, is an empirical question which will be discussed in Section 6 and Section 7.

The design principle underlying the CombCode system is not unique to the Oxnard Basin and reflects a recurring feature of groundwater governance. The Mid-Kaweah Subbasin, for example, similarly adopts a pumping allocation system and permits allocation transfers, but limits transfers to within five miles, constraining the spatial scope of coordination. The Madera Subbasin allows allocations to be shared within designated farm units, creating a comparable internal flexibility for operators managing contiguous parcels. These examples suggest that the findings of this paper may generalize to other SGMA settings, and that effective groundwater market design requires careful attention to how the regulated entity is defined.

3 Data

The empirical analysis integrates well-level groundwater extraction, farm boundaries, crop-specific land use, and regulatory allocations into a field-level panel spanning 2014-2023. This section describes the primary data sources and the construction of key variables.

3.1 Groundwater Extraction and Operational Units

The foundation of the dataset consists of well-level groundwater extraction records provided by the FCGMA, covering 1983-2024. Records contain semi-annual extraction volumes self-reported by well operators and regularly monitored by the FCGMA. Each record identifies the operator name, a Combined Code (CombCode) representing the aggregated reporting unit, and a State Well Number (SWN) serving as a permanent well identifier. Wells sharing a CombCode pool their allocations, and the CombCode structure therefore captures the organizational boundaries within which internal reallocation occurs.

3.2 Field-Level Panel Construction

The primary observational unit is an agricultural field cultivated by grower i in year t. Constructing this unit requires linking well-level extraction records to agricultural land parcels through a two-step geospatial procedure.

First, each well's location is identified using its Well Completion Report (WCR), matched one-to-one with the SWN. Second, the described well location is cross-referenced with Ventura County Tax Assessor parcel maps to identify the agricultural property served by each well and delineate the spatial boundaries of the associated farmland.

Crop coverage data are compiled from two sources. The primary source is the California Department of Water Resources (DWR) crop mapping dataset, drawing on LandIQ imagery available for 2014, 2016, 2018, and 2019-2023. Missing years (2015, 2017) are supplemented using the USDA Cropland Data Layer (CDL). To reduce misclassification, particularly for perennial crops, assignments are cross-validated across adjacent years. The agricultural parcel boundaries are then overlaid with the crop datasets to identify the primary crop cultivated on each field-year.

3.3 Key Variables

Irrigation. Consumptive water use of crop is measured using applied evapotranspiration (ET) estimated from OpenET, a satellite-based platform providing high-resolution monthly ET estimates. Annual applied ET is aggregated from monthly values for each field.

Agricultural revenue. I assume output prices are homogeneous across growers within a crop-year. Average revenue per acre for each crop is calculated from the Ventura County Crop and Livestock Report (2014-2023) and assigned to all fields cultivating that crop in the corresponding year.

Extraction Costs. Groundwater extraction costs are proxied using two complementary measures. The first follows the Ferguson (2025) method, which estimates pumping costs from groundwater depth data obtained from California Department of Water Resources monitoring wells, converting depth to energy requirements and then to dollar costs using local electricity prices. Deeper groundwater implies higher lifting costs, providing a field-relevant measure of spatial and temporal variation in extraction costs. The second measure incorporates the FCGMA's tiered surcharge schedule: operators extracting beyond their allocation face tiered per-unit penalties, which effectively raises the marginal cost of extraction. Together, these two components capture both the physical cost of pumping and the regulatory cost of exceeding allocation.

Groundwater Allocation. Allocations are determined under the FCGMA regulatory framework. Where historical extraction records exist, allocations are calculated by applying the regulatory reduction factor to the historical baseline. Where historical data are unavailable, allocations are assigned using the Irrigation Allowance Index (IAI), based on irrigated acreage and crop-specific

coefficients.

Alternative Water Sources. Access to imported/pipeline water is proxied by whether a field lies within the service boundary of a water delivery district. Formal surface water rights are identified from the California State Water Resources Control Board's Electronic Water Rights Information Management System (eWRIMS).

4 Reduced-Form Evidence

This section provides reduced-form evidence on coalition formation and its consequences. I first document three descriptive patterns in the data that motivate the analysis, then examine the crop characteristics associated with coalition membership, and finally estimate the effect of coalition entry on extraction behavior using an event study design.

4.1 Stylized Facts

Groundwater extraction in the Oxnard Basin is highly concentrated among a small number of operators, and this concentration has grown over time. Figure A1 shows that from 2014 to 2023, the top five coalitions increased their share of total agricultural groundwater extraction from approximately 36 percent to 61 percent, while the average number of wells per coalition rose from 2.7 to 4.1. These patterns indicate substantial and expanding institutional scope for internal reallocation within operator portfolios.

Coalition membership is not static. Figure A2 shows that the set of wells associated with a given coalition changes substantially over time, with wells entering and exiting coalitions across years. This turnover indicates that groundwater reallocation occurs not only through the intensive margin of extraction adjustment within a fixed coalition, but also through the extensive margin of coalition expansion and reorganization.

Coalition composition exhibits systematic crop specialization. Figure A3 shows that many coalitions concentrate the majority of their acreage in a limited set of crops, with high-value perishable crops, particularly strawberries and truck crops (a large proportion of truck crops is celery), accounting for a disproportionate share of acreage within the largest coalitions. This pattern is consistent with the hypothesis that supply chain relationships, rather than geographic proximity alone, drive the formation of coordination groups.

4.2 Coalition Formation and Crop Attributes

The stylized facts suggest that crop specialization in groundwater coalitions. To examine this relationship more formally, I estimate a probability model:

$$Y_{ikt} = \alpha + \sum_{s=2}^{4} \beta_s \cdot \mathbf{1}[\text{Perishability}_{ikt} = s] + \gamma \cdot \mathbf{X}_{ikt} + \lambda_t + \varepsilon_{ikt}, \tag{1}$$

where I estimate the outcome Y_{ikt} as (1) an indicator variable of whether field i in coalition k operated in year t belongs to a multi-well coalition, (2) the number of wells in coalition k that field i in year t belongs to, and (3) an indicator variable of whether field i in coalition k operated in year t belongs to a coalition that has well number $N \geq 5$. Perishability is classified into four categories based on whether a crop needs cool chain and shelf life after its harvest, with the least perishable category ($s = 1$) serving as the baseline. The coefficients β_s capture the difference in coalition probability for crops in perishability tier s relative to the least perishable crops. $\mathbf{X}_{ikt}$ is a vector of field-level controls, and λ_t are year fixed effects. Standard errors are clustered at the coalition (k) level.

Table A2 presents the results. Fields cultivating perishable crops are significantly more likely to belong to a multi-well coalition, with the most perishable category being more likely to belong to a larger multi-well coalition. This result is consistent with the supply chain coordination hypothesis, that growers producing time-sensitive crops, who are more likely to operate in vertical coordination system, are also more likely to form a groundwater coalition.

To assess whether coalitions are internally homogeneous in crop composition, which would further support the supply chain interpretation, I conduct a permutation test for within-coalition crop concentration. For each coalition k in year t, I compute the acreage share of the dominant crop:

$$CropShare_{kt} = \max_{c} \left(\frac{Acreage_{ckt}}{Acreagekt} \right). \tag{2}$$

Higher values of $CropShare_{kt}$ indicate stronger crop specialization within the coalition. The null hypothesis is that coalition membership is no more crop-heterogeneous than a random grouping of fields drawn from the basin population.

To test this null hypothesis, I employ a permutation test. In each of 1000 permutation draws, fields are randomly reassigned to coalitions while preserving the size distribution of observed coalitions. I compute $CropShare_{kt}$ for each permuted assignment and construct the empirical distribution of this statistic under the null. The observed distribution of $CropShare_{kt}$ is then compared against the permutation distribution. A permutation p-value is calculated as the share of draws in which the permuted mean of $CropShare_{kt}$ equals or exceeds the observed mean. This test is nonparametric and does not rely on distributional assumptions about crop composition across

coalitions.

The permutation test design preserves the number and size of coalitions and the overall crop composition of the basin. Therefore, the test isolates the question of whether the observed degree of within-coalition crop homogeneity is attributable to sorting rather than to the basin-wide distribution of crops. Figure 1 presents the results. Observed within-coalition crop share lies to the right of the permutation distribution, indicating that coalitions are significantly more crop-homogeneous than would be expected by chance. This result suggests that coalition formation reflects organizational choices tied to shared production characteristics.

4.3 Allocation Compliance Inside and Outside Coalitions

The results above show which wells form coalitions, but not whether coalitions use the flexibility that pooled compliance provides. If coalition members pool their allocations and reallocate water among themselves, some members will extract more than their own well-level allocation while others extract less. Wells inside coalitions should therefore comply with their individual allocations less often than wells operating alone, even if the coalition as a whole stays within its aggregate allocation.

I measure compliance as the share of well-years in which a well's extraction does not exceed its own allocation, $x_{it} \leq A_{it}$. For wells in coalitions, I also measure compliance at the level at which the FCGMA evaluates it: the share of well-years in which the coalition's total extraction does not exceed its total allocation, $\sum_{j \in k} x_{jt} \leq \sum_{j \in k} A_{jt}$.

The comparison is consistent with internal reallocation. Single wells comply with their allocation in 78.6 percent of well-years. Wells in coalitions comply with their own well-level allocation in only 59.3 percent of well-years, about 19 percentage points less often. Evaluated at the coalition level, however, the compliance rate of these same wells rises to 78.0 percent, almost identical to that of single wells. Coalition members therefore exceed their individual allocations much more often, but coalitions as a whole stay within their aggregate allocation about as often as wells that cannot pool. This pattern suggests that pooling is used mainly to move water among members rather than to raise the coalition's total extraction.

Figure A4 illustrates this with one coalition of 25 wells. Many member wells extract more than their individual allocation, while others extract well below it. The coalition's aggregate extraction of 2,207.1 acre-feet nevertheless stays below its aggregate allocation of 2,344.6 acre-feet, so the coalition complies even though a substantial share of its wells individually are not.

4.4 Effects of Coalition Entry on Extraction

The perishability and homogeneity results establish that coalition formation is systematically related to crop characteristics, but do not directly address whether coalition membership changes extraction behavior. To estimate the effect of coalition entry on extraction, I exploit within-well variation in coalition status using an event study design centered on the timing of a well's first transition into a coalition.

The identifying variation comes from wells that transition from a single-well to a coalition or change of coalition during the sample period. I construct a balanced panel of event windows centered on each well's transition year E_i, retaining up to three years before and after the transition. The pre-transition period provides a within-well baseline against which post-transition extraction is compared.

Crucially, the FCGMA requires that any change in operator status be approved prior to the reporting period in which it takes effect, ruling out contemporaneous manipulation of coalition status in response to within-year extraction shocks.

I estimate the following regression-discontinuity specification in event time:

$$\log x_{it} = \mu_i + \lambda_t + \alpha \cdot \text{EventTime}_{it} + \beta \cdot \text{Post}_{it} + \rho \cdot (\text{EventTime}_{it} \times \text{Post}_{it}) + \varepsilon_{it}, \tag{3}$$

where $\text{EventTime}_{it} = t - E_i$ is the running variable measuring years relative to the coalition transition year E_i for well i, $\text{Post}_{it} = \mathbf{1}[t \geq E_i]$ is an indicator for the post-transition period, μ_i are well fixed effects absorbing all time-invariant well heterogeneity, and λ_t are year fixed effects capturing common shocks. Standard errors are clustered at the coalition level. The coefficient of interest is β, which estimates the discrete jump in log extraction at the moment of coalition entry, holding the pre-transition trend constant. The interaction term ρ allows the slope of the event-time trend to differ before and after the transition.

Figure 2 presents the results. Pre-entry means are stable and flat for both series, supporting the assumption of no differential pre-trend in extraction prior to coalition entry. Following entry at E_i, extraction increases sharply for both groups, with the single-to-multi series rising from a mean log extraction of approximately 3.2 in the pre-period to above 4.0 at entry. This immediate jump is consistent with the reallocation mechanism upon joining a coalition, a well gains access to the pooled allocation of the broader coalition, relaxing its individual extraction constraint and enabling higher extraction than was feasible as a standalone well. Extraction partially reverts in subsequent years but remains elevated relative to the pre-entry baseline.

Table A3 presents the RD estimates. The pre-transition trend coefficient α is small and statistically insignificant in both specifications, confirming the absence of a differential pre-trend. The post-transition indicator β is large and statistically significant: wells transitioning from single well

to a coalition experience a jump of approximately 1.06 log points in extraction at the moment of entry (column 1), while the broader any-coalition-change sample yields a smaller jump. The interaction term ρ is statistically indistinguishable from zero in both columns, indicating that the post-transition trend does not differ significantly from the pre-transition trend.

4.5 From Patterns to Mechanisms: The Role of a Structural Model

The reduced-form evidence establishes the patterns of coalition that they are formed systematically and membership changed extraction behaviors. They do not, however, answer two questions that are central to the argument of this paper. Both questions require a measure of the marginal value of groundwater at each well, which is not directly observed in the data. I therefore develop a structural model in Section 5 and Section 6 for two purposes.

First, the model is needed to interpret reallocation within coalitions. The empirical evidence shows that the reallocation happens within the coalition, and extraction behaviors changed at the entry of a coalition, but not whether the water moves toward its higher-value users. If coalitions behave like a market, or like a planner maximizing the value of their pooled water, wells that extract beyond their own allocation should be those with the highest marginal value at the allocation. Because the marginal value of groundwater depends jointly on crop revenue, irrigation efficiency, and pumping cost, no single observed observed variable reveals it. The structural model recovers each component, which allows me to test the direction of reallocation and to identify which component drives any departure from the efficiency prediction.

Second, the model is needed to evaluate institutional alternatives that are not observed. With the recovered marginal value functions, I solve for the competitive equilibrium under alternative definitions of trading group, holding the basin-wide allocation fixed. This allows me to quantify how much of the gain from a formal market coalitions could deliver if they behaved under the efficiency-maximizing assumptions, how observed coalition behavior compares with that benchmark, and what a directional trading restriction would cost.

5 Model

5.1 Marginal Value of Groundwater Allocations

I present a simple conceptual framework to characterize the marginal value of groundwater allocations across agricultural producers and to illustrate the sources of heterogeneity in these marginal values.

Consider growers indexed by i producing crops indexed by c in year t within the Oxnard Basin.

Agricultural output per acre, Q_{ict}, depends on irrigation water consumed by the crop, X_{ict}, and a vector of other inputs z_{ict}:

$$Q_{ict} = f_{ct}(X_{ict}, z_{ict}) \tag{4}$$

The production function is non-decreasing in its arguments and is assumed homogeneous across growers cultivating the same crop in the same year, reflecting the relatively uniform soil, climate, and cultivation conditions within the Oxnard Basin. Land use and crop choice are taken as exogenous and fixed within a year.

A key distinction is between irrigation water consumed by crops and groundwater extraction. Let x_{ict} denote groundwater extraction. The relationship between extraction and crop water consumption is mediated by a grower-specific irrigation technology:

$$X_{ict} = \Gamma_{it}(x_{ict}), \quad \Gamma'_{it}(x) > 0, \quad \Gamma''_{it}(x) \leq 0 \tag{5}$$

so that greater extraction increases irrigation at a weakly diminishing rate. The function $\Gamma_{it}(\cdot)$ captures irrigation efficiency, conveyance losses, and the availability of alternative water sources.

Growers face a binding regulatory allocation A_{ict} imposed by the FCGMA. Taking output prices p_{ct} and input prices w_{ict} as given, the grower's problem is:

$$\max_{x_{ict}, z_{ict}} \pi_{ict} = p_{ct} f_{ct}(\Gamma_{it}(x_{ict}), z_{ict}) - C(h_{it}) x_{ict} - z'_{ict} w_{ict} \quad \text{s.t. } x_{ict} \leq A_{ict} \tag{6}$$

where $C(h_{it})$ is the marginal cost of groundwater extraction, increasing in the depth to the water table h_{it}. A natural alternative specification would let the marginal pumping cost depend on aggregate basin-wide or CombCode-wide extraction directly, on the logic that one well's extraction lowers the water table and raises pumping costs for nearby wells, which represents a classic common-pool stock externality. I do not model $C(h_{it})$ this way for the following reason. Under SGMA, the FCGMA's individual well allocations A_{it} are themselves set with reference to the basin's sustainable yield target. Therefore, the allocation constraint is precisely the instrument through which the social planner already internalizes the extraction externality. Modeling pumping costs as a function of aggregate extraction on top of a binding allocation constraint would double-count the externality. In the later counterfactual exercise, I take the allocation as fixed across all counterfactual trading regimes and ask only how reallocation within that cap affects welfare, and the relevant marginal cost object is the private pumping and administrative cost alone.

When the allocation constraint is binding, the shadow value of groundwater is:

$$\lambda_{ict} = \underbrace{p_{ct} f_X(\cdot)}_{\text{(a)}} \cdot \underbrace{\Gamma'_{it}(x_{ict})}_{\text{(b)}} - \underbrace{C(h_{it})}_{\text{(c)}} \geq 0 \tag{7}$$

where (a) is the marginal revenue of irrigation, varying by crop and year; (b) is the marginal conversion of extraction into effective irrigation, varying across growers by irrigation technology and access to alternative water sources; and (c) is the marginal extraction cost, varying across wells by pumping depth. Together, these three components generate substantial heterogeneity in the marginal value of groundwater allocations across growers, even within a small and relatively homogeneous agricultural region.

5.2 Coalition Coordination

The heterogeneity in shadow values documented above creates potential gains from reallocation. A coalition k consists of a set of growers $i \in k$ who coordinate extraction decisions and pool their regulatory allocations, subject only to the aggregate constraint:

$$\sum_{i \in k} x_{ict} \leq \sum_{i \in k} A_{ict} \tag{8}$$

In the absence of coordination costs, a coalition would allocate extraction across members until marginal values are equalized:

$$\lambda_{ict} = \lambda_{jct} \quad \forall i, j \in k \tag{9}$$

Let $\Pi_k^{\text{ind}} = \sum_{i \in k} \pi_{ict}(A_{ict})$ denote aggregate profits under independent operation and Π_k^{coal} denote aggregate profits under coalition coordination. Denoting coordination costs as F_k, a coalition forms if and only if:

$$\Pi_k^{\text{coal}} - \Pi_k^{\text{ind}} \geq F_k \tag{10}$$

This condition implies that reallocation occurs only when heterogeneity in marginal values is sufficiently large relative to coordination costs. The coalition structure of the FCGMA maps directly onto this coalition framework: operators sharing a coalition pool their allocations and can reallocate internally without engaging in the formal market. Section 6 tests whether observed reallocation moves in the direction the model predicts, and Section 7 uses it to bound what coalitions would deliver if they behaved efficiently. Coalitions may also form for reasons unrelated to groundwater, such as coordination in the output market or other agricultural input market, in which case Equation 10 can be violated.

5.3 Counterfactual Market Equilibrium

The counterfactual scenario is a competitive permit market operating at the well level, in which the coalition aggregation of allocations is not permitted. Each well faces the market price independently, so large operators cannot net their allocations internally and trade only the residual. Under

a basin-wide cap equal to aggregate allocation $\bar{A}$, wells trade until marginal values are equalized at the market-clearing price p^*:

$$\sum_i x^*_{ict} = \bar{A} \quad \text{and} \quad \lambda_{ict}(x^*_{ict}) = p^* \quad \forall i \tag{11}$$

This first-best outcome assumes zero transaction costs and full participation. I measure the efficiency gain of each trading regime relative to command-and-control, in which every well extracts exactly its own allocation:

$$\Delta W = \Pi^{\text{regime}} - \Pi^{\text{CAC}}. \tag{12}$$

Comparing the basin-wide market with a regime in which trade is confined to existing coalitions separates the gain that coalitions could deliver under efficient internal reallocation from the gain that only a formal market can deliver. Comparing efficient within-coalition trade with observed extraction then shows how far actual coalition behavior falls short of that benchmark.

6 Structural Estimation

Recovering the shadow value λ_{ict} from Equation 7 requires estimating its three components separately. The recovered shadow values are then used to compute coalition gains and simulate the counterfactual market equilibrium.

6.1 Marginal Revenue of Irrigation

I assume a constant-elasticity production function in irrigation, conditional on other inputs at their optimum:

$$Q_{ct} = A_c X_{ict}^{\beta_c} \tag{13}$$

where Q_{ct} is yield per acre, X_{ict} is irrigation water proxied by evapotranspiration. A_c is a crop-specific productivity shifter, and $\beta_c \in (0,1)$ is the irrigation elasticity of yield, or yield response to irrigation. The marginal revenue of irrigation is then

$$p_{ct} f_X(\cdot) = p_{ct} \beta_c A_c X_{ict}^{\beta_c - 1} = \frac{\beta_c (p_{ct} Q_{ct})}{X_{ict}}. \tag{14}$$

Parameter β_c is the irrigation elasticity of yield by crop. Table A6 lists the values of β_c calibrated by regressing $\log(Q_{ct})$ on $\log(X_{ct})$. $(p_{ct}Q_{ct})/X_{ict}$ is the revenue per irrigation by crop year, which is computed from the Ventura County Crop Report and OpenET evapotranspiration. Table A5 summarizes the crop revenue documented in the Crop Report. Figure A5 shows the distribution of

evapotranspiration over the 1,599 operating well-years: the mean is 2.01 feet, the median is 2.06 feet, and the 5th to 95th percentile range 1.22 to 2.72 feet.

6.2 Marginal Conversion of Extraction into Irrigation

The term $\Gamma'_{it}(x_{ict})$ captures the marginal conversion of one additional acre-foot of groundwater extraction into crop-consumptive irrigation. I specify the irrigation technology as a joint conversion of all water inputs into consumptive water use. Let total water input be the sum of groundwater extraction x_{it} and alternative water sources $W_{it} = W_{it}^{\text{surfacerights}} + W_{it}^{\text{district}}$, where $W_{it}^{\text{surfacerights}}$ is surface water delivered under formal rights recorded in eWRIMS and W_{it}^{district} is district-delivered water[7] proxied as:

$$W_{it}^{\text{district}} = \bar{w}_{dt} \cdot \text{Acreage}_{it} \cdot \mathbf{1}\{i \in d\} \tag{15}$$

where $\bar{w}_{dt}$ is total water delivered by district d in year t divided by total irrigated acreage within the district. Fields outside any district boundary are assigned $W_{it}^{\text{district}} = 0$. The irrigation technology converts total water input into consumptive water use (proxied by applied ET) according to a Cobb-Douglas specification:

$$X_{ict} = \Gamma_{it}(x_{ict}) = \gamma_{it} \cdot (x_{ict} + W_{it})^{\alpha} \tag{16}$$

where $\alpha \in (0,1)$ governs the curvature of the conversion, common across fields, and γ_{it} is a field-year-specific efficiency shifter capturing time-varying differences in irrigation technology, infrastructure, and water delivery conditions. The marginal conversion of groundwater extraction into consumptive water use is then:

$$\Gamma'_{it}(x_{it}) = \alpha \gamma_{it} (x_{it} + W_{it})^{\alpha - 1} \tag{17}$$

which is decreasing in total water input, reflecting diminishing returns to additional water application.

Taking logs of Equation 16 and decomposing the efficiency shifter as $\log \gamma_{it} = \mu_i + \delta_i \cdot t + \nu_{it}$, where μ_i is a field fixed effect, δ_i is a field-specific time trend capturing gradual adoption of efficient irrigation technology, and ν_{it} is an idiosyncratic shock, the estimating equation becomes:

$$\log X_{ict} = \mu_i + \delta_i \cdot t + \alpha \log(x_{ict} + W_{it}) + \varepsilon_{it} \tag{18}$$

The parameter α is identified from within-field, within-trend variation in total water input. The field-year efficiency shifter is then recovered as $\hat{\gamma}_{it} = \exp(\hat{\mu}_i + \hat{\delta}_i \cdot t)$.

[7] I estimate the Pumping Trough Pipeline water deliveries under the United Water Conservation District project.

Table A7 reports α under alternative specifications. The parameter estimation used in the main analysis uses $\hat{\alpha} = 0.044(0.013)$ under the specification of Equation (18). It is positive in every specification that does not include an instrument, ranging from 0.0071 (when well-specific trends are replaced with year FE) to 0.0965 (when no trend is included). Figure A6 shows the distribution of $\hat{\gamma}_{it}$ for the 1,578 well-year with at least three operating observations (because a well-specific trend cannot be identified from fewer observations). The median is 1.93 and the 5th-95th percentile range is 1.23 to 2.50. I decompose the variance by the Shapley-Owen approach of decomposing R^2 from crop, year, and well effects, and the well-specific explains the most variance in this irrigation efficiency parameter estimate.

6.3 Marginal Extraction Costs

Marginal extraction costs translate groundwater depth into pumping costs via engineering relationships. The cost of extracting one acre-foot of groundwater from well i in year t is:

$$C(h_{it}) = \kappa e h_{it} \tag{19}$$

where h_{it} is the depth to the water table obtained from CDWR monitoring wells, e is the energy needed to lift one acre-foot of water by one foot, and κ_t is the local agricultural electricity price in year t, which typically ranges from \$0.32-\$0.38. Following Ferguson (W.P.), I set $\hat{e} = 2.6$ kwh/ac-ft/ft, the midpoint of the 1.4-3.8 range of pumping efficiency estimates. Groundwater depth for each field is assigned using the nearest monitoring well within the basin. Figure A7 shows the distribution over the 1,599 operating well-years. Depth has a median of 79 feet, and cost has a mean of \$82 per acre-foot and a median of \$75. Most variation is across wells (standard deviation of well means \$34) and not over time within a well (\$13).

In addition to physical pumping costs, the FCGMA's tiered surcharge schedule imposes a regulatory component of marginal extraction cost for wells extracting beyond their allocation. The effective marginal cost faced by an over-extracting well is therefore:

$$\tilde{C}(h_{it}, x_{it}, A_{it}) = C(h_{it}) + \tau(x_{it} - A_{it}) \cdot \mathbf{1}\{x_{it} > A_{it}\} \tag{20}$$

where $\tau(\cdot)$ is the applicable surcharge tier, increasing in the degree of over-extraction. Together, these two components capture both the physical and regulatory margins of extraction cost variation across wells and years.

6.4 Recovering Marginal Value Dispersion

With all three components estimated, the marginal value of groundwater for each field-year observation is recovered by substituting the estimated parameters into Equation 7:

$$\hat{\lambda}_{ict} = \frac{\hat{\beta}_c(p_{ct}Q_{ct})}{X_{ict}} \cdot \hat{\alpha}\hat{\gamma}_{it}x_{ict}^{\hat{\alpha}-1} - \hat{C}(h_{ict}) \tag{21}$$

I examine the distribution of the recovered marginal values of groundwater across all wells, as this dispersion is the primitive that determines the potential gains from trade. Figure 3 presents the distribution of recovered marginal values $\hat{\lambda}_{ict}$ at extraction across all field-year observations in the basin. The marginal values exhibit substantial heterogeneity, with a long right tail driven by high-value perishable crop producers facing tight individual allocations. This basin-wide dispersion shows that meaningful gains can be achieved from reallocating groundwater in principle.

I also ask whether the marginal value of groundwater are more alike inside coalition than between wells that are not in the same coalition, and whether any such similarity goes beyond the wells growing similar crops. Table A8 and Figure A8 compare dispersion around the coalition-year mean, for the 305 coalition-years with at least two operating wells (959 well-year observations), with the dispersion of random grouping. I use a permutation test that random grouping reassigns the well-years of a year to coalitions with the same sizes as that year's coalition structure, and keep both the coalition structure and crop composition the same. I draw 3,000 random grouping of them under both design. I report four alternative measures of dispersion: the standard deviation of shadow values of groundwater, the median absolute deviation, the standard deviation of log marginal revenue, and the average within-group coefficient of variation of marginal revenue.

Figure A8 shows that the actual coalitions are less dispersed than random group formed within year on every measure. However, the blue histograms (when controlling both coalition structure and crop composition) are shifted to the left of the grey ones (when controlling only on coalition structure), which means that groups with the actual crop composition are already much less dispersed than groups formed without regard to crop. Most of the narrowing within coalitions therefore disappears once crop is held fixed. Coalition members are more alike than wells in general in their marginal values of groundwater at extraction mainly because they grow similar crops, and little of the narrowing remains beyond that. This pattern informs that there is no clear evidence that the similarity we observe in marginal value of groundwater at extraction can be read as efficiency-improving trade. To learn more about the reallocation mechanism, I will use the allocation as the starting point and ask in which direction water is reallocated. If wells with a higher shadow value at the allocation extract more than their allocation (being the "buyer" of water), as the theory predicts, water moves toward higher-value uses.

6.5 Use Marginal Value to Understand the Coalition Reallocation

In the reduced-form evidence, I show that reallocation happens within the coalition, that there are more frequent violations of well-level allocations for wells within a coalition. However, the reduced-form evidence does not say why water is reallocated. If the coalition works as a market, or as a planner maximizing the value of the coalition's water, water should move from wells where have lower shadow value of groundwater allocation to higher value. The buyer test asks whether the direction of reallocation is consistent with the efficiency theory. The decomposition asks which part of the model explains the gap. Together they inform the mechanism of reallocation within coalitions in reality.

I evaluate the shadow value at the allocation:

$$\hat{\lambda}_{ict}(A_{it}) = \frac{\hat{\beta}_c(p_{ct}Q_{ct})}{X_{ict}} \cdot \hat{\alpha}\hat{\gamma}_{it}(A_{it}+W_{it})^{\hat{\alpha}-1} - \hat{C}(h_{ict}). \tag{22}$$

A buyer is a well that extracts more than its allocation, i.e., $x_{it} > A_{it}$. I compare buyers with other wells in the same coalition-year by regressing the shadow value on the buyer status and a coalition-year fixed effect:

$$\lambda_{it}(A) = \alpha_g(i)t + \beta \cdot \mathbf{1}[x_{it} > A_{it}] + \varepsilon_{it}. \tag{23}$$

Coefficient β captures the average gap between buyers and other wells in the same coalition and year. Standard errors are clustered by coalition. If reallocation moves water toward higher-value uses, $\beta > 0$. The main analysis draws on a sample of all coalition-year in 2014-2019 to get rid of any effects from the formal market post 2020. Table 1 shows the main sample in columns (1)-(3) and varies the years and the within-total restriction in columns (4)-(7). The main sample has 98 coalition-years, with 160 buyer well-years and 174 other well-years. The results suggest that buyers do not have a higher shadow value at the allocation. The gap is -$12.8 (s.e. 8.8, p-value = 0.16) against a mean of $30.6 for the other wells. Varying subsamples gives gaps from -$9.6 to +$3.6, but none is significantly different from zero. The gap is positive in the two subsamples that keep only coalition-years within their total allocation and negative in the others, but the differences are small relative to the standard errors.

Table 2 splits the gap of λ into four components:

- The value of water, $\beta_c p_{ct} Q_{ct}/X_{ict}$. This captures the revenue the crop earns per acre-foot of water demand, scaled by the irrigation elasticity.
- The conversion rate of groundwater extraction to irrigation, γ_{it}. This captures the well-level heterogeneity in irrigation efficiency.

	Main: 2014–2019, all coalitions		Other samples		
	(1)	(2)	(3)	(4)	(5)
	λ	$\log MR$	λ	λ	λ
			2014–2019, within total	All years, all coalitions	All years, within total
Buyer ($X_{it} > A_{it}$)	−12.8	−0.135	2.1	−6.6	3.6
	(8.8)	(0.093)	(12.6)	(7.5)	(12.6)
p-value, permutation test	0.34	–	0.77	0.67	0.44
Mean of other wells	30.6	4.57	29.9	45.6	51.8
Coalition-years	98	98	49	155	77
Clusters (CombCodes)	38	38	29	44	34
Coalition-year fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: Each observation is an operating well-year in a CombCode-year with at least one buyer ($X > A$) and one other well. The shadow value is evaluated at the allocation. Main sample: 2014–2019, all coalitions. Because the shadow value is evaluated at the allocation, the main sample does not require a coalition to stay within its total allocation; "within total" keeps only coalition-years in which the coalition's total extraction does not exceed its total allocation. Columns (4)–(7) vary the years and this restriction. Column (3) replaces $\hat{\gamma}_{it}$ with its crop mean. Standard errors clustered by coalition in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: Marginal value at the allocation: buyers versus other wells in the same coalition-year

- Allocation per acre.
- Pumping cost.

In the main sample, buyers' water value is 4% higher (not significant), which contributes \$0.4 in the shadow value gap. Their conversion efficiency is lower. The log gap in $\hat{\gamma}$ is -0.074 (0.025), which lowers their shadow value by \$7.9. Buyers' allocation per acre is 13% higher in logs (p-value = 0.09), which lowers their shadow values by \$3.4 because there is a diminishing in returns of applied water. Pumping cost is slightly higher for buyers, but not showing significant difference. The conversion-efficiency gap is the only significant piece and contribute the largest in explaining the gap.

Coalitions do reallocate water among their members, but I do not find that they reallocate it toward wells where water has a higher marginal value. If the coalition behaved like a market where agents are facing the profit maximizing problem by choosing groundwater extraction, buyers' shadow value at the allocation would exceed that of the wells they buy from. The point estimate is instead negative, although not significant. The decomposition shows this mainly comes from. First, their is no significant difference in how valuable water is to the crop buyers grow; Second, the pumping costs does not favor buyers in the sample, and there is no sign that water moves toward wells where it is cheaper to lift the groundwater; Finally, it shows that buyers tend to have a lower conversion efficiency, which leads to an overall lower shadow value on average.

The efficiency benchmark used above is defined from the perspective of a planner who allocates groundwater to maximize the sum of well-level profits from water, taking crops and other inputs as given. A coalition need not share this objective. Coalitions in the Oxnard Basin form along vertically coordinated supply chains, in which production is often governed by contracts

	Main	Robustness		
	2014–2019, all coalitions	2014–2019, within total	All years, all coalitions	All years, within total
Panel A: log-point gap in each factor of the marginal revenue of water (buyers minus other wells)				
Water value ($\beta_c\, pQ/X$)	0.037 (0.039)	0.103 (0.078)	−0.089 (0.069)	−0.098 (0.123)
Conversion efficiency ($\hat{\gamma}$)	−0.074*** (0.025)	−0.065** (0.030)	−0.063*** (0.020)	−0.064** (0.028)
Allocation per acre ($((A+W)/\text{acre})^{\alpha-1}$)	−0.099 (0.082)	−0.029 (0.107)	−0.018 (0.085)	0.046 (0.084)
Total: $\log MR$	−0.135 (0.093)	0.009 (0.108)	−0.170** (0.078)	−0.115 (0.112)
Panel B: dollars per acre-foot of the gap in λ attributed to each factor				
Gap in λ	−12.8 (8.6)	2.1 (12.7)	−6.6 (7.5)	3.6 (14.0)
Water value	0.4 (4.3)	1.1 (9.2)	−3.7 (7.8)	−0.4 (16.2)
Conversion efficiency	−7.9 (2.6)	−8.0 (4.4)	−9.1 (2.9)	−12.2 (5.5)
Allocation per acre	−3.4 (8.2)	1.8 (13.1)	4.7 (8.1)	8.2 (11.2)
Pumping cost	−2.3 (4.4)	7.3 (3.9)	0.0 (4.8)	4.9 (4.1)
Coalition-years	98	49	155	77
Well-years	334	141	603	284

Notes: The marginal value is evaluated at the allocation. The water value $\beta_c pQ/X$ is the crop-year revenue per acre-foot of applied ET scaled by the irrigation elasticity of yield. The three log-point gaps in Panel A add up to the gap in $\log MR$, where $MR = \lambda + C$. Panel A: coalition-year fixed-effects regressions of each log factor on the buyer indicator ($X > A$), standard errors clustered by coalition. Panel B: the gap in λ minus the gap after setting one factor to its coalition-year geometric mean (arithmetic mean for pumping cost); the factors interact, so the pieces need not sum to the total. Bootstrap standard errors over coalitions (500 draws) in parentheses. Main sample as in Table 1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: What explains the buyer gap in the marginal value at the allocation

between growers and packer-shippers that specify the quantity, timing, and quality of output to be delivered. Highly perishable fresh and specialty crops, such as berries, celery, and leafy greens, are especially likely to be produced under such arrangements (Mighell and Jones, 1963). For a coalition organized around these contracts, a reliable volume of production from each field may matter mroe than the marginal value of groundwater at that field.

This objective could explain the direction of reallocation documented above. Suppose a coalition must deliver a committed quantity of output from each field. Water is then allocated to the fields where it is needed to meet that commitment, not to the fields where its marginal value is the highest. A field with lower irrigation efficiency needs more water to produce the same output, so it would extract beyond its allocation, while more efficient fields would extract less and supply the difference. This is consistent with the decomposition in Table 2, where the only significant difference between buyers and other wells in the same coalition is the buyers' lower conversion efficiency. Under this interpretation, the observed reallocation is not inefficient from the coalition's point of view. It is efficient with respect to an objective that includes the value of meeting supply commitments, which the groundwater model does not capture.

This interpretation remains a hypothesis. Testing it would require data on other agricultural inputs, field-level outputs, and the terms of grower-shipper contracts, which are unobservable here, and other explanations, such as allocating water jointly with other inputs or keeping year-to-year use patterns stable, cannot be ruled out. The distinctions matters for interpreting the results that follow. The comparison of observed and efficient allocation behavior in Section 7 measures the

cost of observed behavior in terms of profit from groundwater alone. If coalitions value supply reliability, part of the gap reflects a benefit the model does not count.

7 Counterfactual Trading Regimes

The structural estimates in the previous section do not support the idea that coalitions move groundwater toward its higher-value uses. Coalition members reallocate water among themselves, but the wells that extract more than their allocation do not have a higher shadow value at the allocation than the wells they share a coalition with. Shadow values at extraction are more alike inside coalitions than between random wells, but that similarity mostly comes from members growing similar crops. Therefore, I cannot take as given that coalition formation raises the value generated by the groundwater. Coalitions do not appear to operate as the substitute for a formal market in reality.

The coalition structure still matters for the counterfactual. Suppose that coalitions did behave as the theory predicts, so that members trade until marginal values are equal within each coalition. Trade is then limited to members of the same coalition, and the dispersion in marginal values across coalitions and other wells is left unexploited. A formal market, on the other hand, allows trade across the whole basin and equalizes marginal values everywhere. The gains from the formal market are therefore the gains from equalizing within coalitions plus the gains from equalizing across them. How large that part is depends on how much of the dispersion in marginal values lies within coalitions and how much between them.

Therefore, in this section, I compare and contrast three different counterfactual trading regimes

7.1 Setup

Using the recovered shadow value function $\hat{\lambda}_i(x)$ from Equation 21, I solve for counterfactual reallocation under alternative definitions of who is permitted to trade with whom.

A trading group G is a set of wells permitted to reallocate extraction among themselves subject to a shared aggregate allocation cap. The competitive equilibrium within G is the unique price p_G^* solving

$$\sum x_i^* = \sum A_i, \;\; \hat{\lambda}_i(x_i^*) = p_G^* \;\; \forall i \in G, \tag{24}$$

where A_i is well i's regulatory allocation. Inverting Equation 21 gives $x_i(p) = K_i/(p+\hat{C}(h_i))^{1/(1-\hat{\alpha})}$, where $K_i = \hat{mr}_{ci} \cdot \hat{\alpha} \cdot \hat{\gamma}_i$ captures the crop-specific marginal revenue term, the irrigation conversion curvature and field-year efficiency shifter. Equation 24 is then solved numerically for p_G^* within each group.

Aggregate profit associated with an extraction path x follows directly from integrating Equation 21, $\pi_i(x) = \int_0^x \hat{\lambda}_i(u)du = K_i x^{\hat{\alpha}}/\hat{\alpha} - \hat{C}(h_i)x$, which is used to evaluate welfare under each

counterfactual regime.

7.2 Baseline and Trading Regimes

I focus on 2020, the first year of formal trading under the FCGMA. The sample is the $N = 151$ wells that are operating in 2020 (water applied per irrigated acre of at least 10% of the crop requirement), spanning 87 CombCodes. Of these, 93 wells belong to multi-well coalitions and 58 are outside coalitions.

The baseline for welfare comparison is command-and-control: every well i extracts exactly its individually assigned allocation A_i and no reallocation is allowed. The baseline and each counterfactual trading regime share the same 2020 allocation, $\bar{A} = \sum_i A_{i,2020} = 23{,}283$ acre-feet.

I consider three definitions of the trading group G in Equation 24:

1. **Full basin-wide trade.** G is the entire basin. Every well can trade with every other well. This is the frictionless well-level market benchmark.

2. **Within-coalition trade only.** G is each CombCode separately. Wells reallocate only among members of their existing coalition. Wells without a partner have no one to trade with and stay at their allocation.

3. **Hybrid.** Multi-well CombCodes clear internally within their coalition boundaries. All wells outside coalitions are additionally pooled into one common market among themselves. Coalitions do not participate in this outer market, but individual growers who are not part of a coalition can now trade with one another.

Regime (2) approximates the ceiling on what a fully efficient CombCode system with zero internal coordination costs could already deliver on its own. Regime (1) is the first-best benchmark of a formal, basin-wide market. Regime (3) isolates how much of the gap between (1) and (2) is attributable specifically to coalitions withholding themselves from formal exchange, creating some sort of "trading barriers" in the market.

7.3 Welfare Decomposition

Table 3 reports the aggregate welfare gain of each trading regime relative to command-and-control.

Within-coalition reallocation alone (regime 2) captures 26 percent of the gain a frictionless well-level market (regime 1) could deliver. Opening the market to non-coalition wells while leaving coalitions untouched (regime 3) raises the captured share to 63 percent. The remaining 37 percent could only be materialized if coalitions themselves also participate in formal exchange, rather than confining reallocation to their own internal boundary.

	(1) Full basin trade	(2) Within-coalition only	(3) Hybrid
ΔW vs. command-and-control	\$814,311	\$214,645	\$512,772
Share of full-market gain	100%	26%	63%
Buyers / sellers	70 / 80	41 / 46	70 / 75
Total acre-feet reallocated	8,167	3,142	6,489

Notes: ΔW is the change in aggregate profit relative to command-and-control, in which every well extracts exactly its own 2020 allocation. The sample is the 151 operating wells of 2020 in 87 CombCodes (93 wells in multi-well coalitions, 58 outside coalitions). All regimes hold aggregate extraction in each trading group equal to the group's aggregate allocation; the aggregate allocation is 23,283 acre-feet. A buyer (seller) extracts more (less) than its allocation by more than one acre-foot. Acre-feet reallocated are the sum of the positive net trades.

Table 3: Counterfactual welfare gain by trading regime, 2020

The gaps between regimes are informative about where the residual gains of a formal market lie.

- \$599,666 (regime 1 minus regime 2): the gain a well-level market adds beyond what existing coalitions could capture on their own.
- \$298,126 (regime 3 minus regime 2): the gain realized simply by giving non-coalition growers access to a market.
- \$301,539 (regime 1 minus regime 3): the additional gain that requires coalitions, which could capture their internal reallocation gains, to also become active participants in the formal market.

Reallocation is also more extensive at the margin under less restrictive regimes: 70 wells are net buyers and 80 net sellers under full basin-wide trade, versus 41 buyers and 46 sellers when trade is confined within existing coalitions. The remaining wells have no eligible trading partners and do not participate at all. Market-clearing prices differ across groups: \$40 per acre-foot in the basin-wide market, \$22 in the pool of wells outside coalitions, and a median of \$48 across the 28 CombCodes that have at least two wells in the sample.

This decomposition gives an upper bound on what the pre-market institution can deliver. Even if coalitions equalized marginal values perfectly within their boundaries, they would capture only about a quarter of the gain from a frictionless market. Giving wells outside coalitions access to a market would add another 37 percent of the full-market gain, and the remaining 37 percent could be realized only if coalition wells traded in the formal market as individual wells. Section 7.4 shows that observed coalition behavior falls short even of this upper bound. A market design that allows coalitions to keep pooling allocations internally therefore forgoes a large share of the potential gain, even under the most favorable assumption about how coalitions behave.

7.4 Compare Counterfactual with Observation

The regimes above show what coalitions could deliver if wells inside them traded until marginal values were equal. The structural estimates suggest that observed reallocation does not follow that rule, so this subsection compares the two directly. I restrict the sample to the 93 operating wells that belong to coalition in 2020 and evaluate welfare, $\sum_i \pi_i(x_i)$, under four scenarios for how these wells extract. In the first, each well extracts its allocation. In the second, each well extracts what it actually extracted in 2020's observation. In the third, each coalition extracts its aggregate allocation, split across its wells in proportion to observed extraction, which keeps the observed pattern across wells but removes the difference in total volume. In the fourth, wells trade within their coalition until shadow values are equal, which is regime (2) above. The first scenario is the command-and-control benchmark for these wells, and the fourth is the benchmark that the theory of informal coordination under efficiency-improving incentives predicts.

	Extraction (AF)	Profit	ΔW vs. (1)	% change vs. (1)	Wells above / below allocation	Within-coalition s.d. of λ
(1) Allocation	14,099	\$27,894,845	–	–	0 / 0	86
(2) Observed extraction	10,921	\$27,165,668	−\$729,177	−2.61%	26 / 66	394
(3) Observed shares, scaled to allocation	14,099	\$27,496,038	−\$398,808	−1.43%	40 / 52	219
(4) First-best trade	14,099	\$28,109,490	\$214,645	0.77%	41 / 46	49

Notes: The sample is the 93 operating wells in multi-well CombCodes in 2020; wells outside coalitions are left out. Scenario (1): every well extracts its allocation. (2): every well extracts what it actually extracted in 2020. (3): each coalition extracts its aggregate allocation, split across its wells in proportion to observed extraction. (4): wells trade within their coalition until shadow values are equal (regime 2), with the coalition's aggregate extraction equal to its aggregate allocation. Profit is the model-implied profit $\sum_i \pi_i(x_i)$. A well is above (below) its allocation if it extracts more (less) than its allocation by more than one acre-foot. The last column is the root of the average within-coalition variance of the shadow value at the scenario's extraction, in dollars per acre-foot.

Table 4: Welfare of coalition wells under four scenarios, 2020

Table 4 reports the results. Observed behavior is worse than following the allocation. Under observed extraction, coalition wells earn \$729,177 less than under the allocation, a fall of 2.61 percent of profit, and under the observed pattern scaled to the full allocation they earn \$398,808 less (1.43 percent). First-best trade within coalitions, in contrast, adds \$214,645 (0.77 percent). The distance between observed behavior and the theory benchmark is therefore \$943,822 in the second scenario and \$613,453 in the third, several times larger than the gain the benchmark itself delivers.

The change in the dispersion of shadow values points in the same direction. Within coalitions, the standard deviation of the shadow value is \$86 per acre-foot when wells extract their allocation and falls to \$49 under first-best trade, but it rises to \$394 under observed extraction and to \$219 under the scaled pattern. Observed extraction thus spreads the marginal value of water within coalitions further apart than the allocation does, and does not bring it together as the theory predicts. This is consistent with the buyers test in Section 6, which find that wells extracting more than their allocation do not have higher shadow values than their partners.

These numbers are model-implied and should be interpreted carefully. The loss from extracting less than the allocation depends on the shape of the recovered shadow value function at low extraction, where it is steep, and the model leaves out constraints that may bind in practice. It also counts only profits from groundwater. If coalitions reallocate water to meet output commitments to packer shippers (Section 6.5) , the value of meeting those commitments is not included. The comparison therefore likely overstates the cost of observed behavior in dollar terms. What it does show is the direction: coalitions reallocate water, but the reallocation moves marginal values to a more dispersed direction.

7.5 Where Does the Water Go? Crop Composition

Because the shadow value $\hat{\lambda}_i$ is driven in part by crop-specific marginal revenue, the counterfactual reallocation is expected to track crop type closely. Table 5 aggregates each well's net counterfactual trade by crop and reports the net acre-feet bought, and Table 6 reports the net acre-feet sold, under each of the three regimes. Flowers and nursery, peppers, greenhouse crops and strawberries are the net buyers under full basin-wide trade, and citrus (1,404 acre-feet), dry beans, cole crops, turf, grain and hay, avocados and truck crops are the net sellers.

Net buyers	(1) Full basin trade	(3) Hybrid	(2) Within-coalition
Flowers and Nursery	1,659	926	0
Peppers	607	78	78
Greenhouse	600	650	0
Strawberries	224	443	260

Notes: Net acre-feet traded relative to the crop's aggregate allocation. Positive values indicate that the crop is a net buyer under that regime. This table only shows crops with net trade of at least 100 acre-feet in absolute value under full basin trade.

Table 5: Net counterfactual acre-feet purchased by crop, 2020

The within-coalition column is usually the smallest in magnitude, and for several crops (flowers and nursery, greenhouse, dry beans, grain and hay) collapses to zero. This is because within-coalition trade can only reallocate water among the specific crops that happen to be co-located within an existing coalition, and Section 4 showed that coalitions are crop-homogeneous (in 48 percent of coalition-years with at least two operating wells, all wells grow a single crop). A crop that is not well represented inside multi-well coalitions has essentially no coalition partners to trade with, even if its basin-wide shadow value is high. The direction of a crop's trade within coalitions also need not match its direction in the basin-wide market, because the coalition's clearing price reflects only its own members: truck crops are a net buyer within coalitions (+584) and a net seller basin-wide (-130).

Net sellers	(1) Full basin trade	(3) Hybrid	(2) Within-coalition
Citrus	−1,404	−1,374	−78
Beans (Dry)	−456	−221	0
Cole Crops	−355	−107	−47
Turf Farms	−292	−282	−310
Grain and Hay	−199	−180	0
Avocados	−158	−158	−39
Truck Crops	−130	148	584

Notes: Negative values indicate that the crop is a net seller (extracts less than its aggregate allocation) under that regime. This table only shows crops with net trade of at least 100 acre-feet in absolute value under full basin trade.

Table 6: Net counterfactual acre-feet sold by crop, 2020

7.6 Spatial Patterns and Directional Trading

Figure A9 maps every operating well in 2020, colored by its net counterfactual trade under full basin-wide trade (regime 1). Blue marks a net buyer (extracting more than its individual allocation), red a net seller, with color depth and marker size both scaling with the reallocated volume.

Buyers concentrate in the western part of the basin along the coast and in the south, while sellers are located in the north-central part and in the eastern part of the basin. The wells in the PTP area, which receive district water, are net sellers (3,987 acre-feet in total across 67 wells), since district water already lowers their marginal value of groundwater. Most of the buying, however, takes place outside the two Special Protection Areas defined below: wells inside them account for 1,640 of the 8,167 acre-feet bought under full basin-wide trade (20 percent).

Figure A10 and Figure A12 repeat the exercise for regimes (2) and (3). The underlying pattern of who would like to buy or sell is similar across the three maps, but a visibly larger share of wells are locked out of any transaction altogether under regime (2). Regime (3) recovers most of that lost activity by letting non-coalition wells trade with each other, even though coalitions themselves remain confined to internal reallocation.

Geography plausibly matters for the consequences of that trade, and here the two institutional channels differ importantly. A formal market operating under FCGMA oversight can, in principle, be designed with considerations on protecting vulnerable areas by putting limitations on water buying in zones prone to seawater intrusion. The FCGMA's centralized, double-blinded trading platform is, by construction, a natural point at which such directional constraints could be enforced, since every formal transaction passes through agency review.

Internal reallocation through the CombCode system faces no comparable constraint. Because compliance is evaluated at the aggregate operator level rather than the individual well level, an operator whose portfolio spans both an inland well and a coastal well can freely shift extraction toward the coastal well as long as the CombCode's aggregate extraction stays within its aggregate

allocation, even if that reallocation increases groundwater extraction in an area already at elevated risk of saline intrusion.

This asymmetry suggests informal coordination may create external costs that a formal market could have avoided by design. I take a first step toward quantifying this concern by constructing an explicit directional trading restriction.

The two Special Protection Areas (SPAs) are defined in the FCGMA Groundwater Sustainability Plan (GSP). Wells are not permitted to be net buyers of water within the SPAs. One SPA is the coastal zone, the area of the Oxnard Plain near the coast from Point Mugu north to the Santa Clara River, where the GSP documents saltwater intrusion. The other is the Pumping Trough Pipeline (PTP) management area along the eastern basin.[8] Of the $N = 151$ operating 2020 wells, 4 (1,091 acre-feet of allocation) fall within the coastal zone and 67 (8,481 acre-feet) within the PTP zone. Within each SPA, a well's counterfactual extraction is constrained to $x_i \leq A_i$: it may sell water (extract less than its individual allocation) or hold at A_i, but it may not buy.

Applying this constraint to regime (2), within-coalition trade alone, shows how much of today's informal reallocation would already be foreclosed by such a rule. Under unrestricted within-coalition trade, wells in the two SPAs collectively buy 1,002 acre-feet, 32 percent of the 3,142 acre-feet reallocated within coalitions in that regime. Banning this leaves non-SPA members of the same coalitions to extract 585 acre-feet more than before, and the rest cannot be absorbed within the coalition (171 acre-feet occurs in coalitions composed entirely of SPA-zone wells, where the corresponding seller is equally constrained and reverts to its own allocation). About one-third of the reallocation that the coalition system could permit informally would be foreclosed under a formally enforced directional restriction.

I also repeat the full basin-wide trade counterfactual (regime 1) with the SPA restriction imposed. Table 7 reports the resulting welfare cost. Restricting directional trade lowers the gain from a frictionless, basin-wide market from \$814,311 to \$752,295 relative to command-and-control, a loss of \$62,016 or 7.6 percent of the unrestricted gain. Put differently, a formal market that enforces this single directional rule still recovers 92.4 percent of the efficiency gain a fully unconstrained well-level market would deliver. The hybrid regime keeps \$443,724 of its \$512,772 gain, and regime (2) keeps \$179,059 of its \$214,645.

That a single, easily monitored directional rule retains more than nine-tenths of the frictionless market's efficiency gain suggests the trade-off between protecting vulnerable sub-areas and cap-

[8] These boundaries are approximations for this exercise rather than an official FCGMA management-area GIS layer. The coastal zone is the land within a uniform inland distance of the shoreline between the Point Mugu headland and the mouth of the Santa Clara River, with the distance chosen so that the area equals the 16 square miles reported in the GSP; the true boundary follows chloride contours. The PTP zone is the area within one kilometer of the pipeline loop, digitized from the district's facilities map, whose georeferencing error is about one kilometer. They are intended to be illustrative of the general institutional point rather than definitive delineations.

	Value
Wells in the coastal zone	4
Wells in the PTP zone	67
Allocation in the two zones (AF)	9,572
Within-coalition trade (regime 2)	
SPA buying under unrestricted trade (AF)	1,002
Total within-coalition buying (AF)	3,142
Share of reallocation blocked (%)	31.9
Full basin-wide trade (regime 1)	
ΔW, unrestricted	\$814,311
ΔW, with SPA no-buy restriction	\$752,295
Welfare cost of the restriction	\$62,016
Welfare cost (% of unrestricted gain)	7.6
Share of unrestricted gain retained (%)	92.4

Notes: Wells in a Special Protection Area (SPA) may not be net buyers: $x_i \leq A_i$. The coastal zone is a strip of 16 square miles along the shoreline between Point Mugu and the Santa Clara River, and the PTP zone is the area within one kilometer of the Pumping Trough Pipeline (PTP) loop. AF denotes acre-feet. ΔW is measured relative to command-and-control, as in Table 3.

Table 7: Effect of the SPA directional trading restriction, 2020

turing reallocation gains is not especially large, at least for the two zones considered here. The internal reallocation within the coalition, by contrast, does not address such external costs because an operator cannot be prevented from shifting extraction toward a coastal or PTP well it controls.

8 Discussion

This paper uses the pre-market allocation institution of the first groundwater market under SGMA to inform the design of that market. Under the FCGMA's coalition (CombCode) system, operators controlling multiple wells pool their allocations and reallocate water internally, creating what the groundwater management agency members describe as "private markets" outside the formal platform. These coalitions have become more concentrated over time and are organized along vertically coordinated, crop-homogeneous supply chains. They do reallocate water: the compliance rate is lower at the well-level for members of coalitions, and a well's extraction rises sharply when it joins a coalition. But the reallocation does not follow the marginal value of groundwater. Wells that extract beyond their allocation do not have higher marginal values than their coalition partners, and in the model, observed extraction spreads marginal values within coalitions further apart than the allocation itself does. Even under the favorable assumption that coalitions equalized marginal values among their members, they would capture only 26 percent of the gain from a frictionless

well-level market, because their members grow similar crops.

The findings carry three implications for groundwater market design under SGMA. First, a market should be evaluated against the institution it replaces. Thin trading on a formal platform is sometimes read as a sign that reallocation is already happening elsewhere. In Oxnard, reallocation is happening elsewhere, but it does not deliver the efficiency gains a market would, so low trading volume should not be taken as evidence that the market is unnecessary. Second, the definition of the trading entity is an important design choice. Allowing compliance to be evaluated for groups of wells lets users reallocate within groups that formed for other reasons and whose members have similar marginal values, which limits the gains available inside the coalitions and reduces the incentive to trade outside it. Defining the trading entity at the well level, or requiring that transfers between wells in the same group pass through the market, will achieve higher efficiency gains. Third, the formal market can be used as a tool for supervision and addressing external costs. Because the coalition system evaluates compliance only at the operator level, it cannot restrict where within an operator's portfolio extraction increases. A formal market can impose a directional rule at the point of trade, and the no-purchase rule retains 92 percent of the market's efficiency gain.

The analysis has several limitations that point to extensions. First, the paper shows that coalition reallocation does not follow the marginal value of groundwater, but it does not identify what it does follow. One plausible explanation is that coalitions organized around grower-shipper contracts allocate water to meet committed output from each field, which would send water toward less efficient fields. Distinguishing this from other explanations would require data on other agricultural inputs and contract terms. Distinguishing among them would require data on contracts or field-level input use. Second, the welfare comparison between observed and efficient coalition behavior is model-implied. It depends on the shape of the recovered marginal value function at low extraction levels and omits constraints that may bind in practice, so it likely overstates the dollar cost of observed behavior; its direction can be interpreted more robustly. Third, the counterfactual market assumes zero transaction costs, which bias the estimated market gains upward. Jointly modeling coalition formation and market participation, and recovering the coordination and transaction costs that rationalize observed behavior, would provide a tighter bound on the net gains from market implementation.

Figures

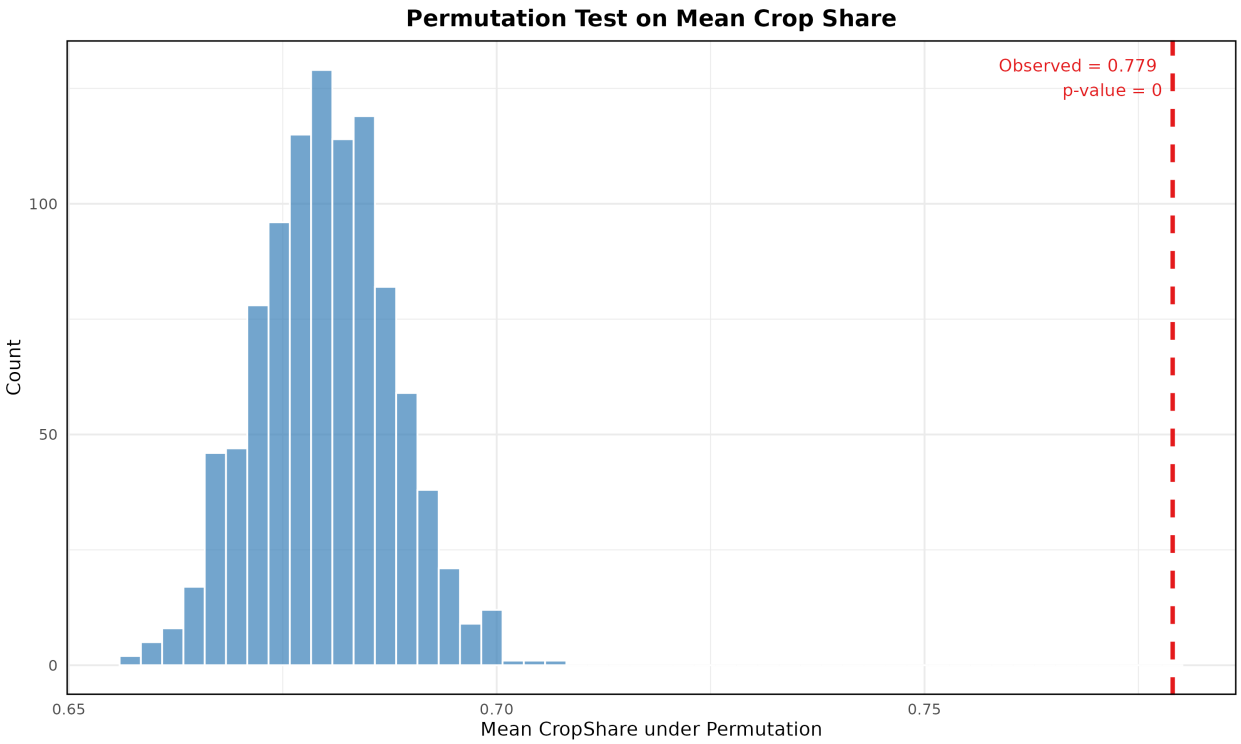


Figure 1: Permutation Test for Within-Coalition Crop Homogeneity. Observed crop share values compared to null distribution from 1,000 random reassignments.

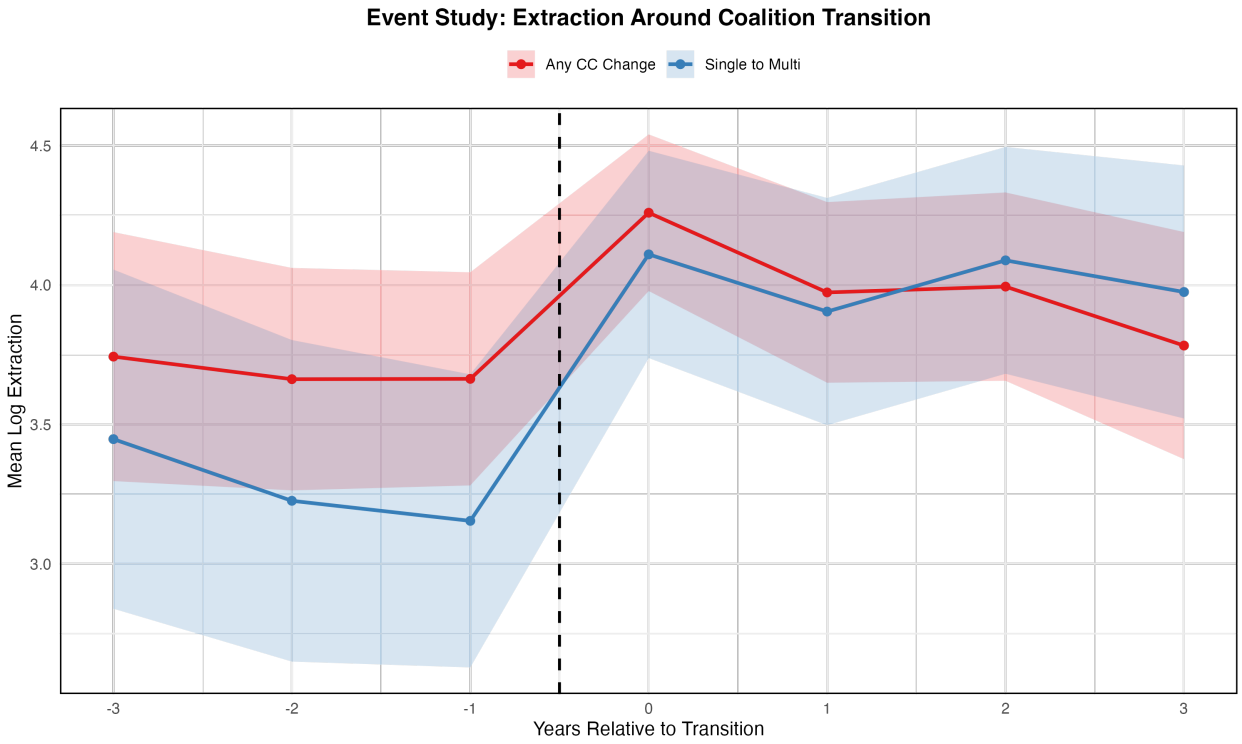


Figure 2: Mean log groundwater extraction around coalition transition, plotted separately for any CombCode change (red) and single-to-multi-well transitions (blue). The dashed vertical line marks the transition year. Shaded bands denote 95% confidence intervals.

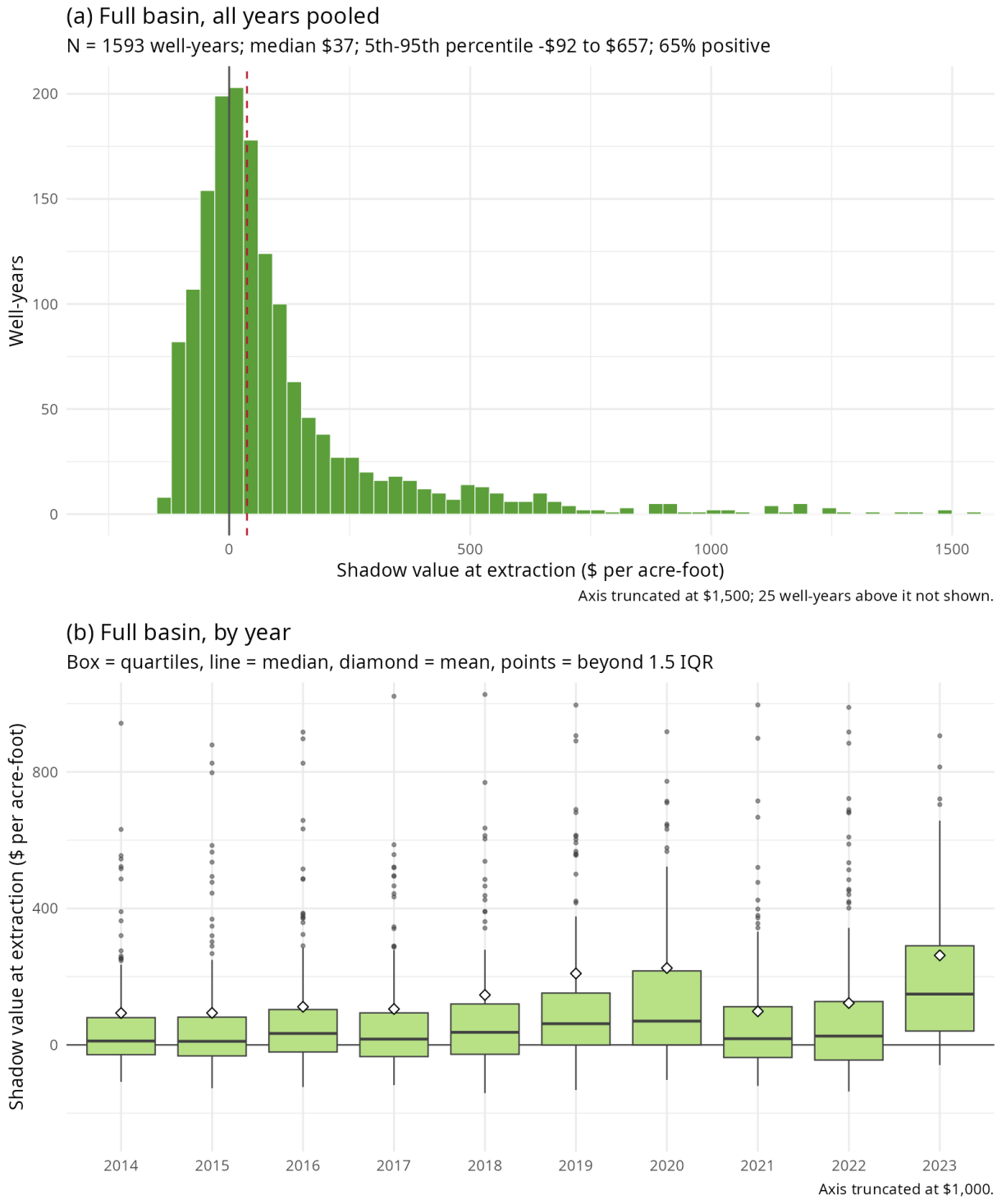


Figure 3: Panel (a) shows the distribution of recovered marginal values of groundwater at the observed extraction pooling all field-year observations in the basin. Panel (b) shows the distribution by year.

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Appendix

	Full Sample		Single-Well		Multi-Well	
Variable	Mean	SD	Mean	SD	Mean	SD
Panel A: Field and Operator Characteristics						
Irrigated acreage (acres)	100.20	82.01	88.47	77.89	110.83	84.21
Multi-well CombCode (%)	0.595	0.491	0	0	1	1
Wells per CombCode†	3.82	4.99	1	0	5.74	5.72
Panel B: Crop and Revenue Variables						
Revenue per acre ($1,000/acre)	44.9	24.1	44.2	24.4	45.5	23.9
Perishability (1-4, where 4 is the most Perishable)	2.62	0.68	2.52	0.70	2.71	0.65
CropShare within CombCode†					0.779	0.221
Panel C: Groundwater and Irrigation Variables						
Groundwater extraction (AF/year)	134.91	172.94	82.08	121.26	182.75	197.20
Groundwater allocation (AF/year)	180.20	141.88	179.93	151.82	180.45	132.32
Crop water consumption (AF/year)	208.87	186.46	179.72	170.96	235.27	195.84
Extraction $\leq$ Allocation (%)	69.4	46.1	83.8	36.8	56.3	49.6
CombCode extraction $\leq$ CombCode allocation (%)	80.9	39.4	83.8	36.8	78.1	41.4
Observations (field-years)	1860		884		976	

Notes: The unit of observation is a field-year. The sample covers agricultural fields in the Oxnard Basin from 2014 to 2023. Groundwater extraction and allocation data are from the Fox Canyon Groundwater Management Agency (FCGMA). Crop water consumption is proxied by applied ET which is estimated from OpenET and PRISM data. Revenue per acre is measured at the crop-year level using the Ventura County Crop and Livestock Report. AF = acre-feet.

† CombCode-level variables are computed at the CombCode-year level (N = [CombCode-year]); all other variables are at the field-year level.

Table A1: Descriptive Statistics

Dependent Variables:	coalition		well count	large coalition	
Model:	(1)	(2)	(3)	(4)	(5)
	Logit	OLS	OLS	Logit	OLS
Variables					
Ultra-Perishable	3.895***	0.6950***	4.367***	3.516***	0.6164***
	(1.268)	(0.1234)	(1.481)	(0.8836)	(0.1595)
Highly Perishable	1.626***	0.3684***	1.313	0.5356	0.0654
	(0.5128)	(0.1016)	(1.515)	(0.5985)	(0.0762)
Moderately Perishable	1.488***	0.3390***	0.7981	0.6365	0.0771
	(0.5040)	(0.1022)	(1.042)	(0.5759)	(0.0715)
log(acre)	0.5117**	0.1145***	-0.0807	0.1503	0.0191
	(0.2083)	(0.0438)	(0.4241)	(0.1922)	(0.0258)
Fixed-effects					
year	Yes	Yes	Yes	Yes	Yes
Fit statistics					
Observations	1,635	1,635	974	1,635	1,635
Squared Correlation	0.10213	0.09804	0.16364	0.15886	0.15595
Pseudo R^2	0.07886	0.07296	0.02826	0.15110	0.17899
BIC	2,135.9	2,247.2	6,080.4	1,434.3	1,375.1

Clustered (CombCode) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A2: Crop Type and Coalition Formation

Dependent Variable:	log(extraction)	
	Single to Multi	Any CC Change
Model:	(1)	(2)
Variables		
event time	-0.1304	-0.0356
	(0.1364)	(0.0922)
post	1.058***	0.5753**
	(0.3044)	(0.2218)
I(event time × post)	0.0873	-0.1021
	(0.1419)	(0.1139)
Fixed-effects		
well	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	476	766
R^2	0.63173	0.62766
Within R^2	0.05187	0.02177

Clustered (CombCode) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table A3: Regression Discontinuity Results



Figure A1: The top panel shows the concentration of groundwater extraction among top CombCodes, where the dashed line suggests that the unique number of CombCodes has declines over time. The left-bottom panel shows that the maximum number of wells in the CombCode has increased over time, and the right- bottom panel shows that the mean number of wells in the CombCode has increased over time as well. Taken together, this figure suggests that the groundwater extraction has been more concentrated from 2014 to 2023 in the Oxnard Basin.

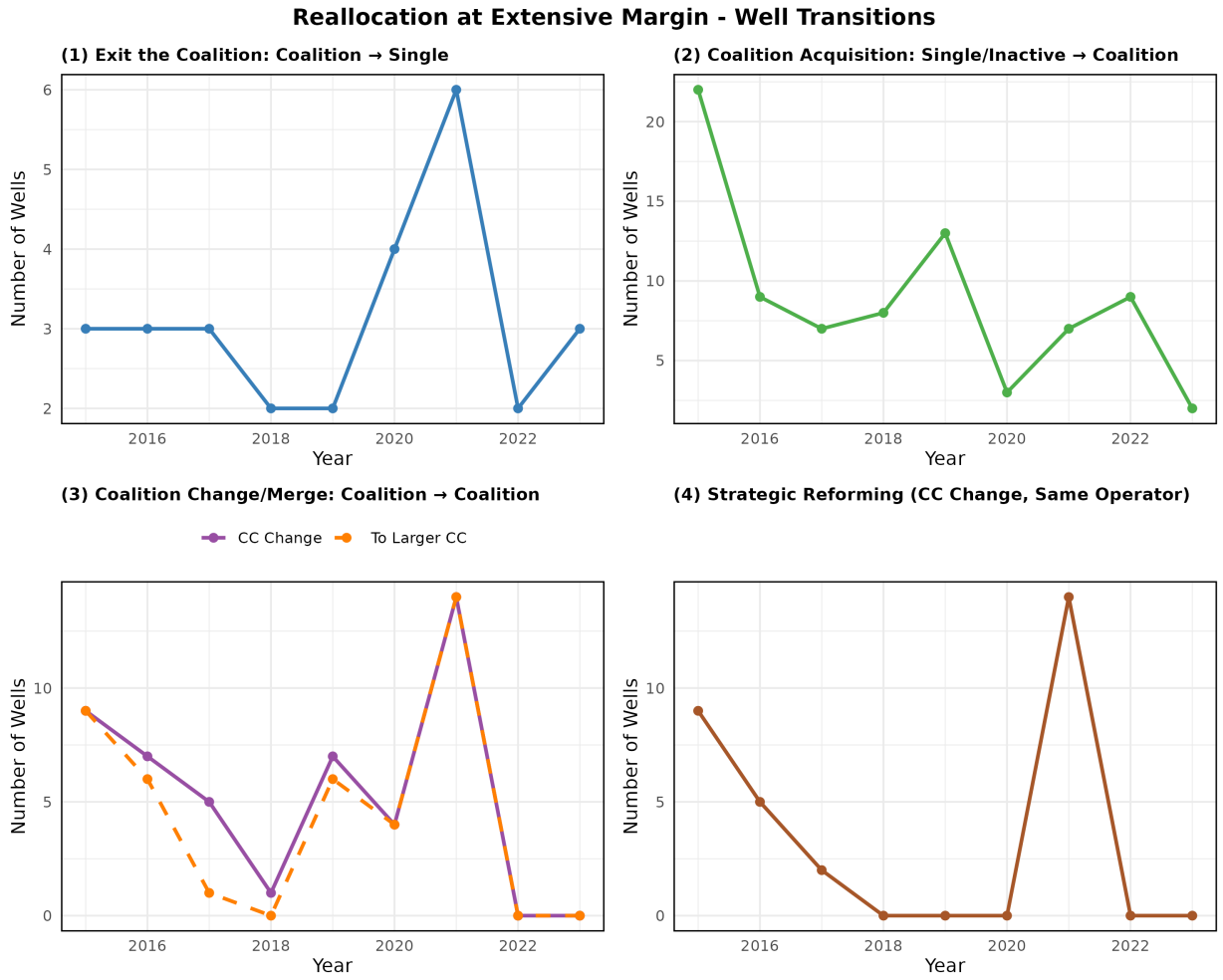


Figure A2: This figure shows the changes in CombCode membership over time. Panel (1) suggests that there exists exits from multi-operation coalition to single-operation, though the number of wells exiting the coalition is relatively small compared to the active wells each year (about 180-190 wells). Panel (2) suggests that frequent coalition acquisition has observed in the basin. Panel (3) suggests that there exists frequent coalition membership changes or coalition merging, especially for small coalition changing to coalition with more wells. Panel (4) further distinguishes the strategic reforming of coalition, i.e., wells changing the CombCodes but not the operator name. For example, wells under XXX-1 and XXX-2 combined to a CombCode called XXX-combined. It suggests that the coalition merge in 2019 is purely dominated by this strategic reforming.

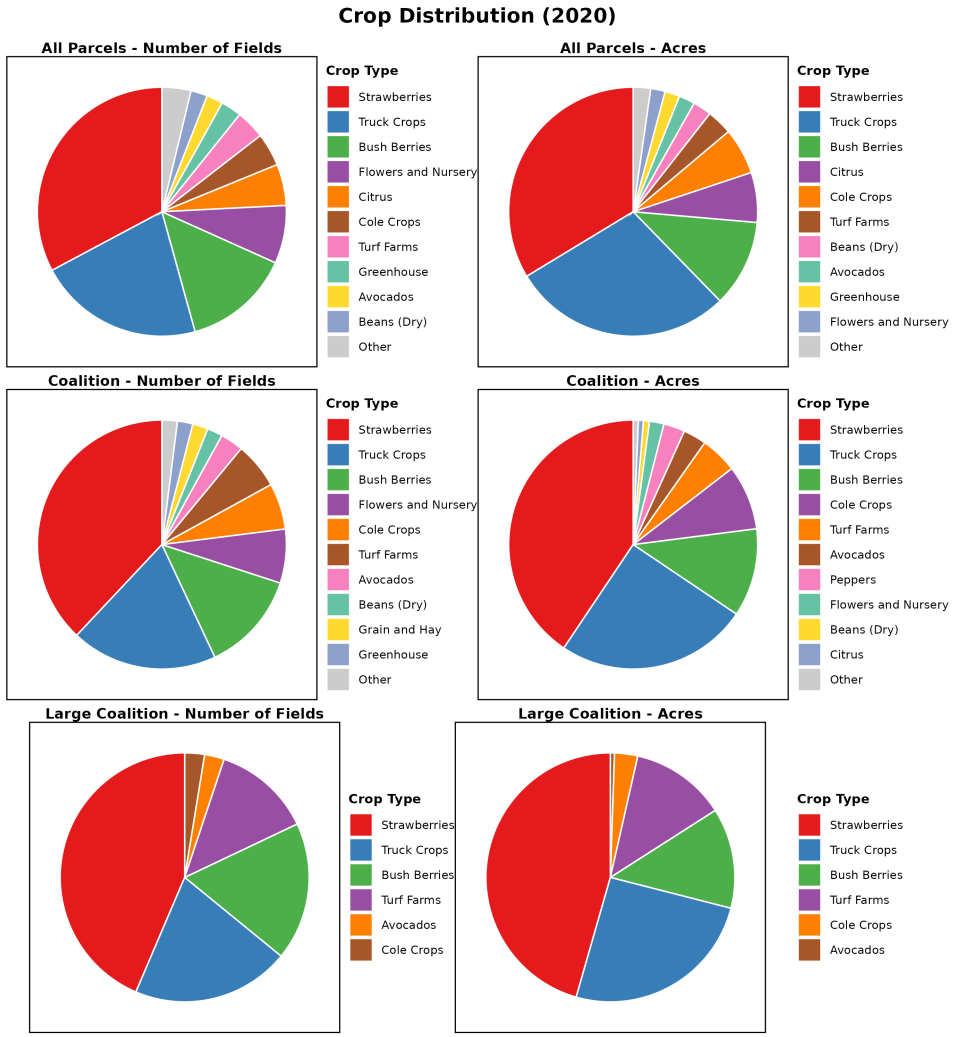


Figure A3: This figure gives one year example of crop composition in the basin (top row), in multi-well coalitions (mid row), and in large coalitions (with well number $N \geq 5$) (bottom row). It shows that coalition composition exhibits some extent of crop specialization, especially for larger coalitions.

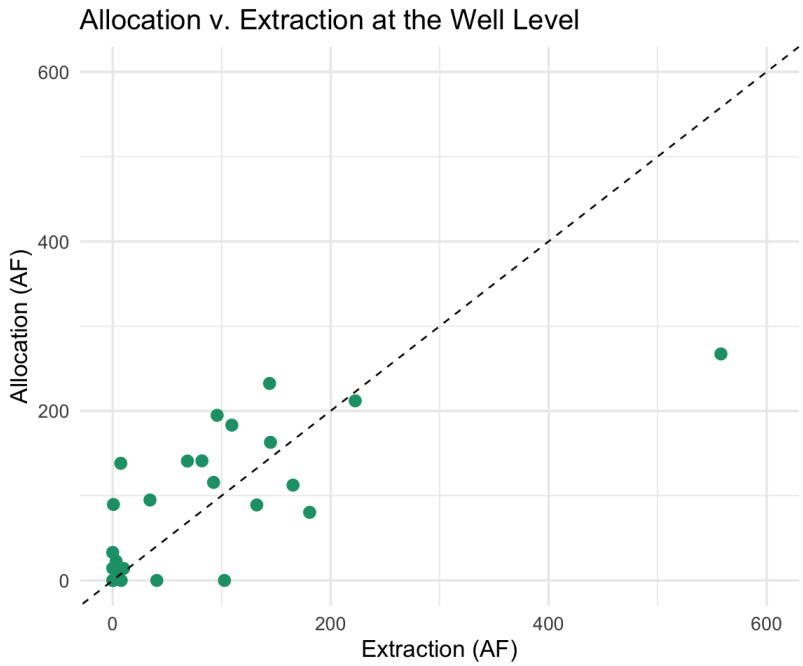


Figure A4: Allocation versus extraction at the well level for one coalition. Each point is a member well; the dashed line marks extraction equal to allocation. Points below the line are wells extracting more than their own allocation. The coalition has 25 wells, an aggregate allocation of 2,344.6 acre-feet, and aggregate extraction of 2,207.1 acre-feet.

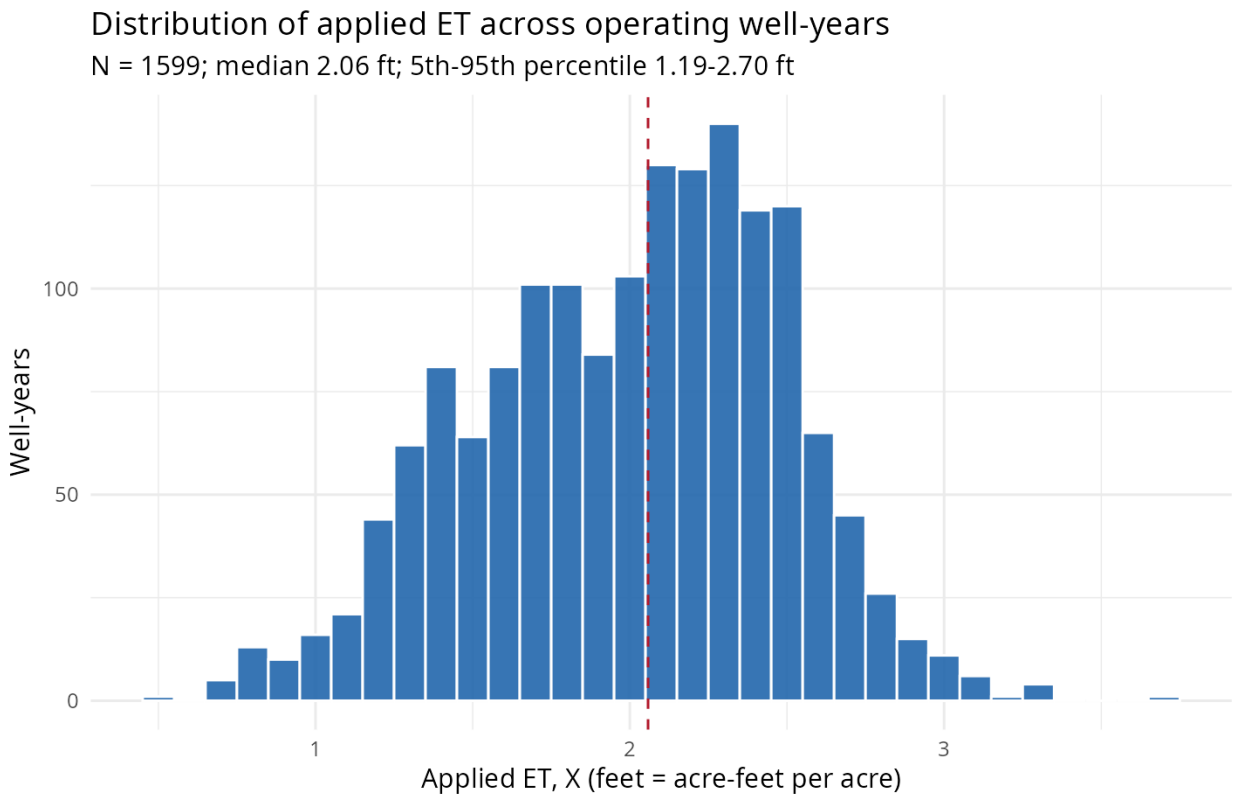


Figure A5: Distribution of Applied Evapotranspiration. The applied ET is measured as depth in feet (equivalently, acre-feet per acre), averaged over the raster pixel in each field, then subtracted by effective precipitation.

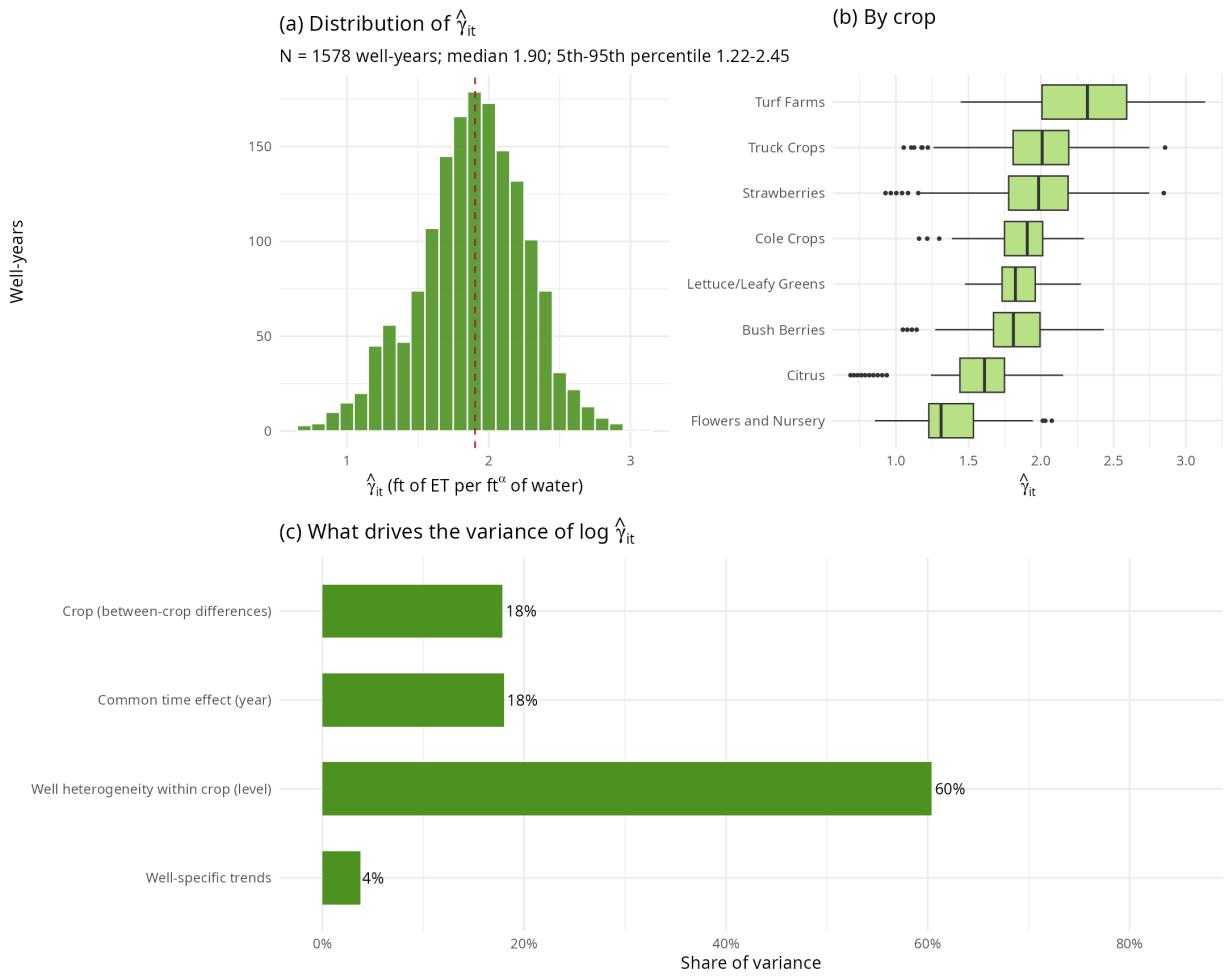


Figure A6: Distribution of Well-Time Specific Irrigation Efficiency Parameter, $\hat{\gamma}_{it}$

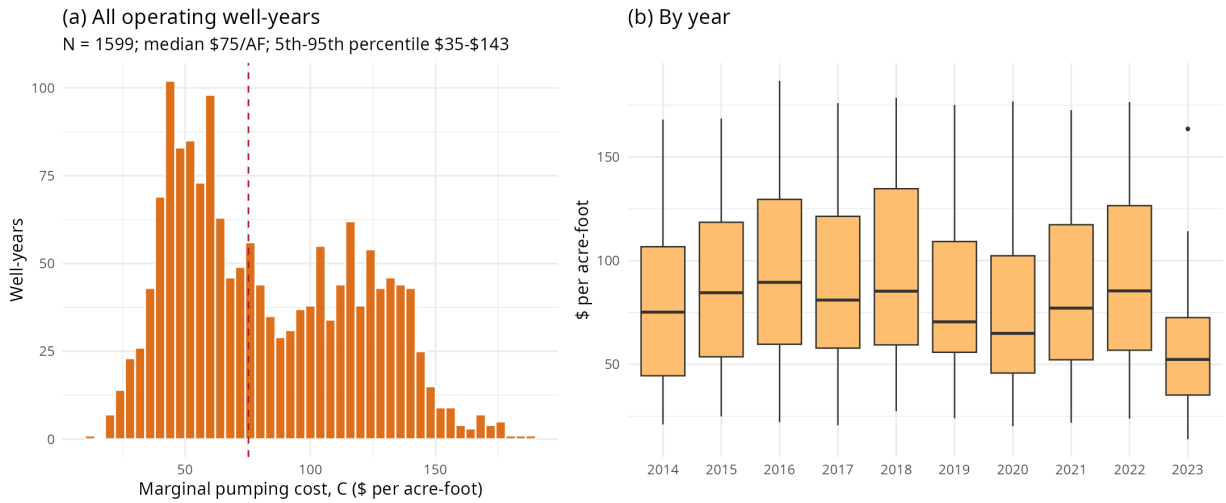


Figure A7: Distribution of Pumping Cost, $\hat{C}(h_{it})$

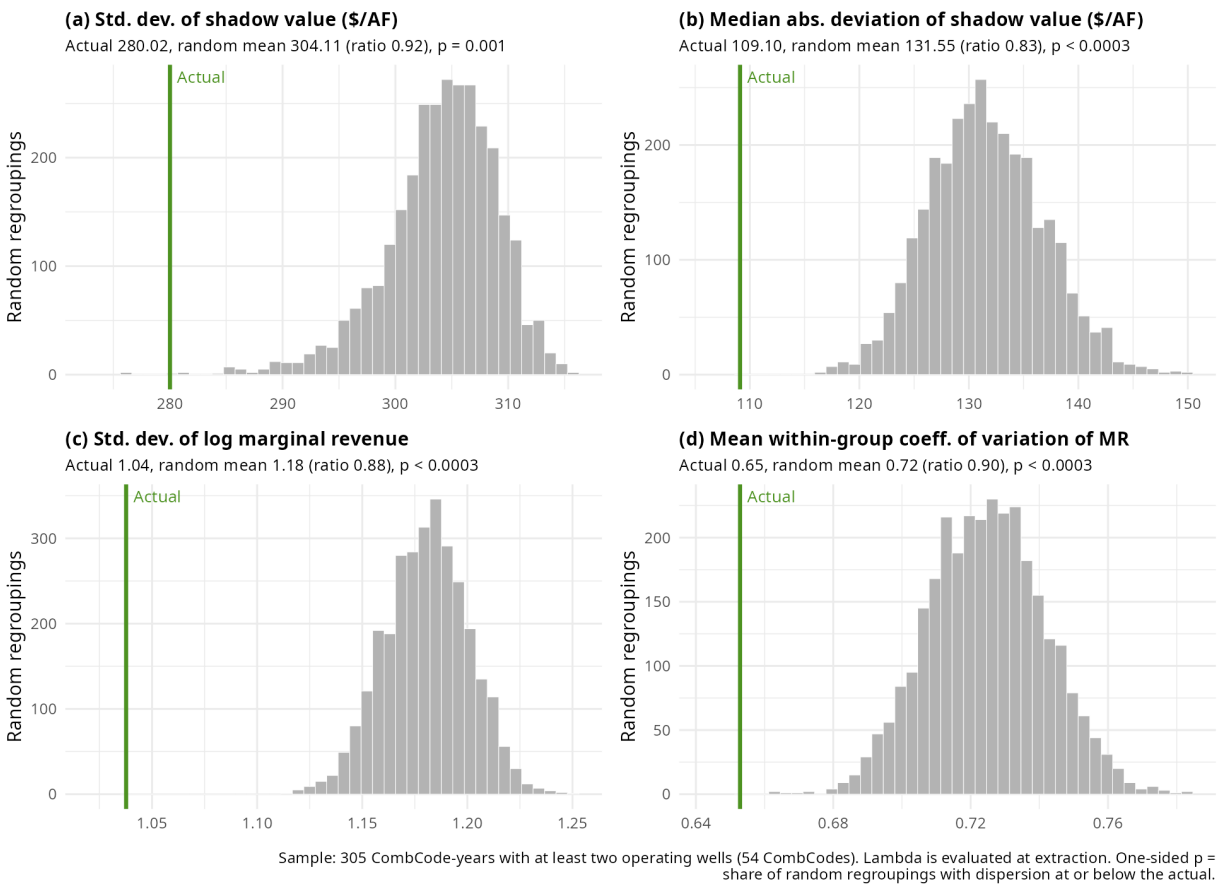


Figure A8: Are shadow values at extraction more alike within coalitions than in random groups?

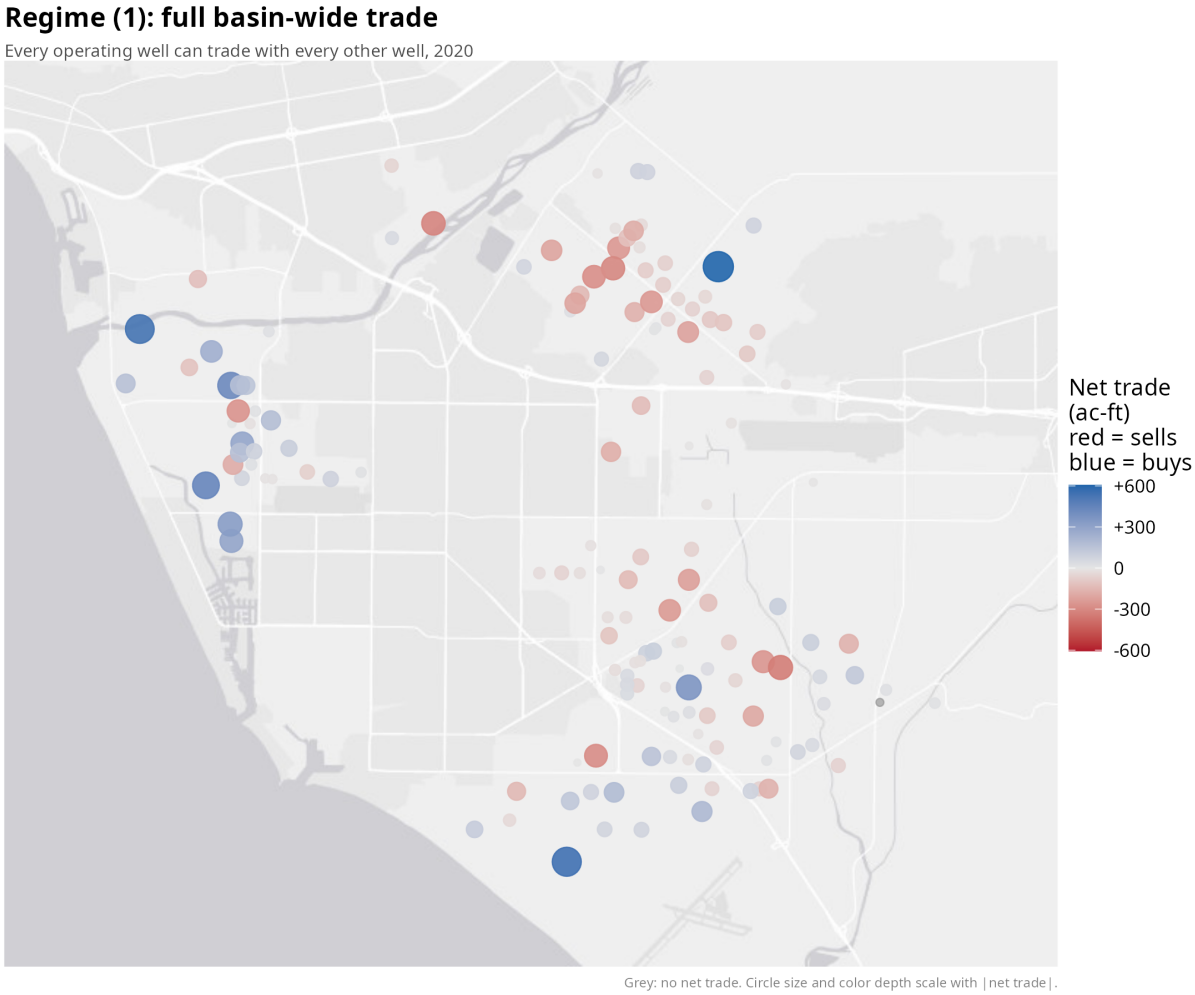


Figure A9: Net counterfactual trade by well under full basin-wide trade (regime 1), 2020. Blue (red) wells are net buyers (sellers) relative to their individual allocation; color depth and marker size scale with the reallocated volume in acre-feet. Grey wells show no net trade.

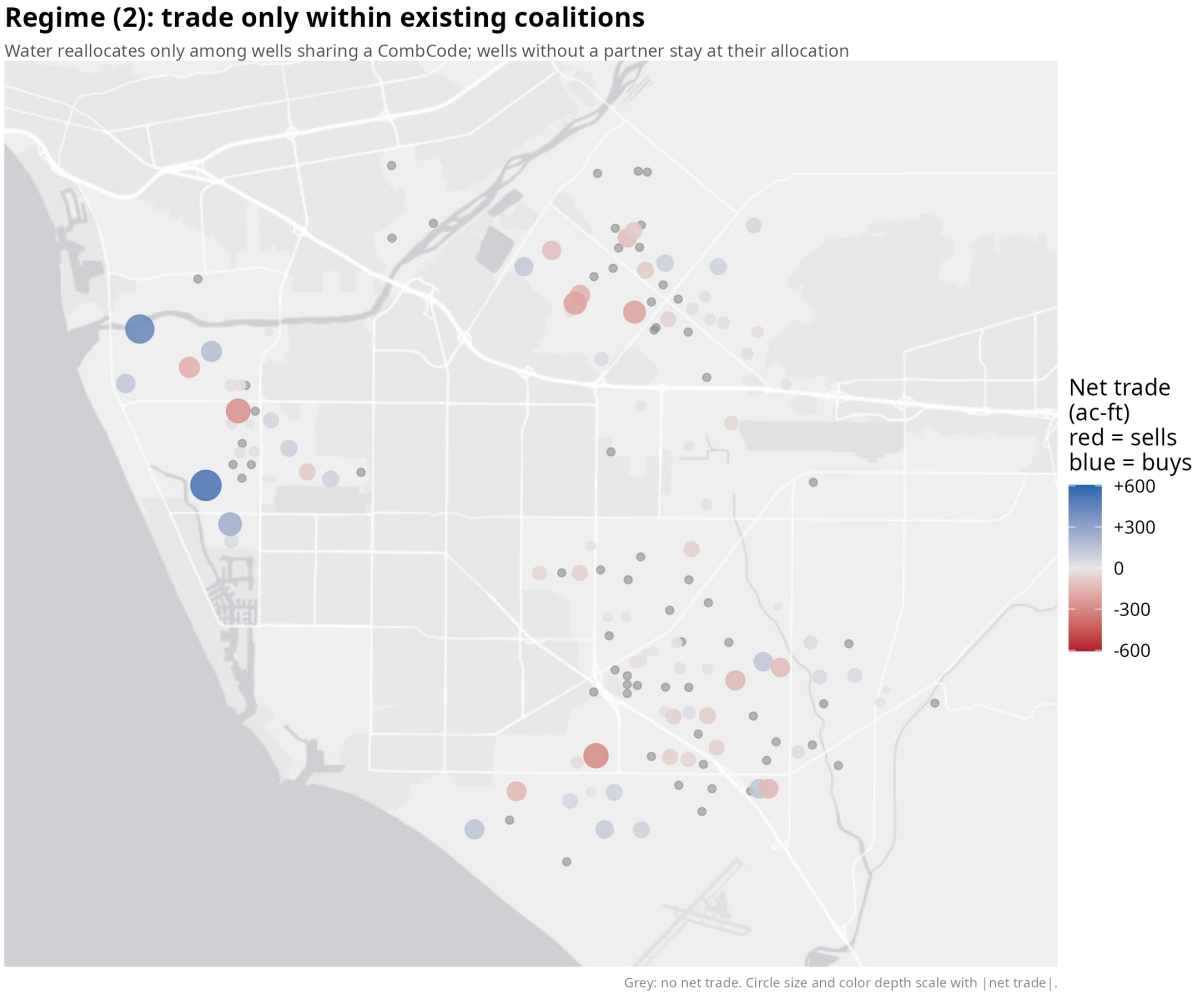


Figure A10: Net counterfactual trade by well under within-coalition trade only (regime 2), 2020.

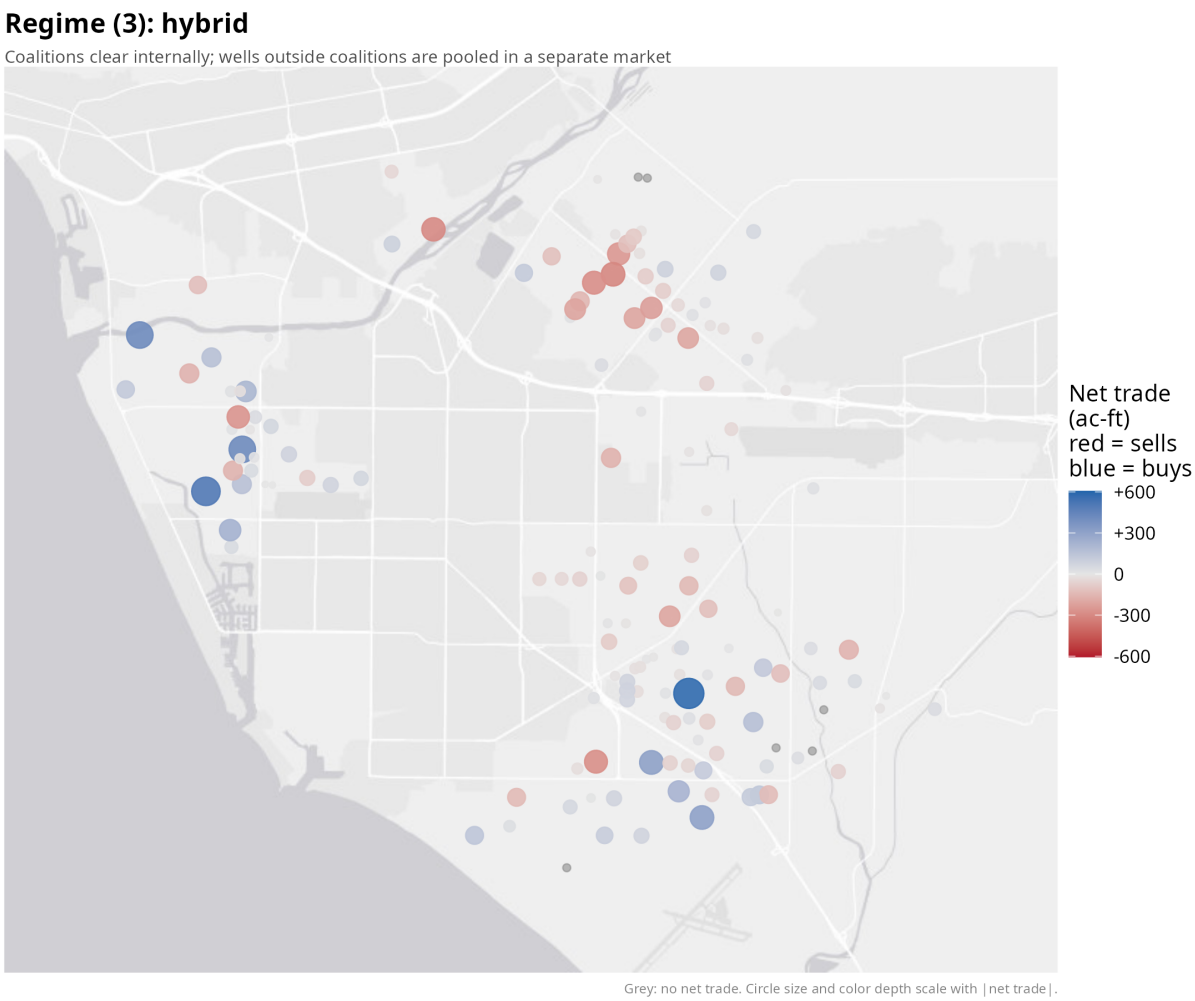


Figure A11: Net counterfactual trade by well under the hybrid regime (3), 2020: coalitions trade internally; non-coalition wells trade in a separate pooled market.

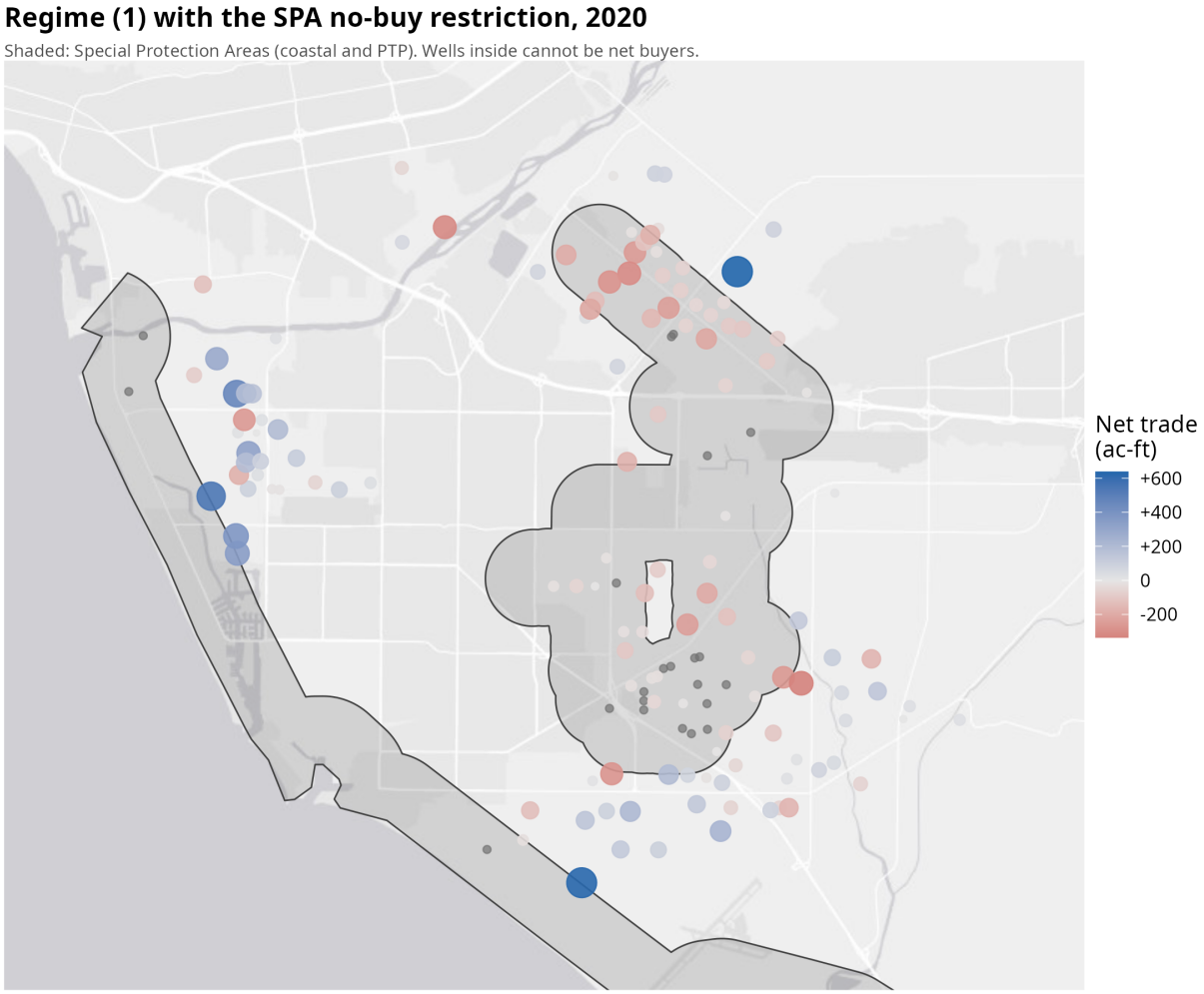


Figure A12: Net counterfactual trade by well under the hybrid regime (3), 2020: coalitions trade internally; non-coalition wells trade in a separate pooled market. Two Special Proctected Areas (SPAs) are drawn: coastal and Pumping Trough Pipeline (PTP). Within the SPAs, no more water can be extracted more than the allocated level at each well.

cc

Crop Category	Variety	Max Shelf Life[a]	Immediate Cold Chain[b]
Avocados	—	12–18 days	yes
Beans (Dry)	—	3–10 years	no
Bush Berries	Blackberries	2–3 days	yes
	Blueberries	1–2 weeks	yes
	Raspberries	2–3 days	yes
Citrus	Lemon	10–12 weeks	yes
	Mandarins	4–6 weeks	yes
	Oranges	8–12 weeks	yes
	Grapefruit	6–10 weeks	yes
Cole Crops	Broccoli	10–14 days	yes
	Cabbage	3–6 weeks	yes
	Kale	2–3 weeks	yes
Field Crops	—	6–12 months	no
Flowers and Nursery	—	3–14 days	yes
Grain and Hay	—	1–3 years	no
Greenhouse	—	4–14 days	yes
Lettuce/Leafy Greens	—	2–3 weeks	yes
Melons, Squash & Cucumbers	Cantaloupe	5–14 days	yes
	Cucumbers	10–14 days	yes
	Squash	1–2 weeks	yes
Peppers	—	2–3 weeks	yes
Strawberries	—	5–7 days	yes
Tomatoes	—	4–7 days	yes
Truck Crops	Carrots	3–4 weeks	yes
	Celery	2–3 weeks	yes
	Onions	1–8 months	no
	Potatoes	4–5 months	no
Turf Farms	—	12–24 hours	yes
Young Perennials	—	n.a	n.a.

[a] *Max Shelf Life* refers to the maximum recommended storage duration under optimal conditions (appropriate temperature, humidity, and atmosphere). Actual shelf life may be considerably shorter under field or transport conditions. Ranges reflect variation across varieties and storage methods.

[b] *Immediate Cold Chain* indicates whether refrigerated storage or transport is recommended immediately after harvest to preserve quality and safety or whether the crop can be stored and transported at ambient temperature without significant quality loss.

Source: University of Tennessee Extension, *Storage of Fresh Produce* (SP768-F). Lado et el. (2019). *Citrus*. Lieu et al. (2024). *A Review of Preservation Approaches for Extending Avocado Fruit Shelf-Life*. Khumalo et al. (2023). *Inefficiency in Land-Side Cold-Chain Logistics*.

Table A4: Crop Perishability by Shelf Life and Cold Chain Requirement

Crop	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Avocados	6493	9703	6978	6826	6032	7094	10936	7673	14442	7407
Beans (Dry)	—	—	—	—	—	—	2866	2047	2178	5962
Bush Berries	51293	46399	38970	39965	43617	48755	51752	49135	55055	61276
Citrus	15190	14387	15572	15439	15335	12668	12241	12270	10452	10299
Cole Crops	—	8328	9573	9809	9941	11176	12048	12902	15145	9060
Field Crops	—	—	—	—	—	185	—	—	—	—
Flowers and Nursery	64694	65130	71491	74707	80069	75168	60458	60968	56955	54785
Grain and Hay	—	—	—	—	—	3480	517	750	775	—
Greenhouse	—	65130	63643	62668	62378	60279	61547	69642	61798	70138
Lettuce/Leafy Greens	—	—	16094	16528	23241	24105	20198	18019	12023	17509
Melons, Squash & Cucumbers	31760	—	13591	—	—	—	—	—	—	—
Peppers	15457	16635	17597	15393	—	16461	22251	—	26217	17133
Strawberries	53995	54860	64018	64048	73632	58521	65376	75514	64572	70492
Truck Crops	14081	14756	14903	14840	14973	15988	13184	14394	12859	12378
Turf Farms	34043	45901	45574	46312	47551	46238	50076	57265	55976	68775
Young Perennial	—	—	—	—	0	—	—	—	—	—

Note: Revenue values are in dollars per acre (rounded to the nearest dollar). — indicates the crop did not appear in the data in that year.
Source: The Ventura County Crop & Livestock Report, 2014-2023.[a] The crop revenue is calculated from dividing the total revenue by the total acre of a certain crop type.

Table A5: County-Level Crop Revenue, 2014-2023 ($/Acre)

[a] https://awm.venturacounty.gov/crop-reports/

Crop	β_c	Operating well-years
Strawberries	0.120 (0.362)	648
Truck Crops	0.400 (0.175)	319
Flowers and Nursery	0.350 (0.274)	139
Bush Berries	0.209 (0.091)	128
Turf Farms	0.150 (0.114)	76
Citrus	0.150 (0.094)	59
Cole Crops	0.396 (0.180)	57
Lettuce/Leafy Greens	0.442 (0.502)	30
Avocados	0.150 (0.191)	20
Greenhouse	0.350	19
Peppers	0.656 (0.562)	13
Beans (Dry)	0.200	8

Notes: β_c is the elasticity of yield with respect to applied evapotranspiration. Values are calibrated by regressing log yield on log evapotranspiration by each crop. Operating well-years are well-years with extraction of at least 1 acre-foot, 2014-2023.

Table A6: Irrigation elasticity of yield, β_c

	(1) Main	(2) No trend	(3) Year FE	(4) No alternative water	(5) Incl. non-operating	(6) IV: depth
$\hat{\alpha}$	0.0440***	0.0965***	0.0071	0.0148**	0.0279***	−1.3965
	(0.0130)	(0.0137)	(0.0071)	(0.0065)	(0.0081)	(4.9171)
Well fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Well-specific trends	Yes	No	No	Yes	Yes	No
Year fixed effects	No	No	Yes	No	No	Yes
Incl. alternative water W	Yes	Yes	Yes	No	Yes	Yes
Extraction ≥ 1 AF	Yes	Yes	Yes	Yes	No	Yes
Observations	1,594	1,594	1,594	1,594	1,691	1,594
Wells	176	176	176	176	182	176
First-stage F						0.1

Notes: Dependent variable is log applied ET per acre. The water input is log of (extraction + alternative water) per parcel acre, in feet. Column (1) is the specification in Equation 18. Standard errors clustered by well in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (6) instruments the water input with log depth to groundwater.

Table A7: Estimates of α

Dispersion measure	Actual coalitions	Random groups, within year			Random groups, within year and crop		
		Mean	Ratio	p	Mean	Ratio	p
Std. dev. of λ ($/AF)	280	304	0.92	0.001	274	1.02	0.777
Median abs. deviation of λ ($/AF)	109	132	0.83	< 0.001	117	0.93	0.031
Std. dev. of log MR	1.04	1.18	0.88	< 0.001	1.07	0.97	0.040
Mean within-group CV of MR	0.65	0.72	0.90	< 0.001	0.67	0.97	0.079

Notes: Sample is the 959 well-years in the 305 CombCode-years with at least two operating wells. Dispersion is measured around the group mean (rescaled by $\sqrt{n/(n-1)}$ for group size) and pooled across groups, except the last row, which is the average across groups of the standard deviation of marginal revenue divided by its mean. λ can be negative with a mean near zero, so a coefficient of variation is computed on marginal revenue (MR $= \lambda + C > 0$) and not on λ. Random groups keep each year's group sizes and are formed within year; the second placebo also keeps each group's crop composition (wells are shuffled only among wells growing the same crop in that year). Ratio is actual over random mean. p is the share of 3,000 random regroupings with dispersion at or below the actual value.

Table A8: Are shadow values at extraction more alike within coalitions than in random groups?